

Cardiac Fibroblasts regulate myocardium and coronary vasculature development via the collagen signaling pathway

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eLife Assessment

The study presents extensive gene expression profiling and bioinformatic analyses, offering insights into the roles of fibroblasts in cardiac development. The large volume of scRNA-seq data is both **compelling** and **important** to the scientific community. All three reviewers agree that the revised manuscript represents a significant improvement and addresses most, if not all, of their previous concerns. The reviewers also acknowledge that detailed mechanistic studies on how fibroblast-derived collagen regulates myocardial and coronary vasculature development are beyond the scope of the current study.

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Abstract

The fibroblast (FB), cardiomyocyte (CM), and vascular endothelial cell (Vas_EC) are the three major cell types in the heart, yet their relationships during development are largely unexplored. To address this gap, we employed RNA staining of the FB marker gene *Col1a1* together with the CM marker gene *Actn2* and the Vas_EC marker gene *Cdh5* at different stages. This approach enabled us to discern the anatomical pattern of cardiac FBs and identify approximately one EC and four CMs directly interacting with each FB. Molecularly, through the analysis of single-cell mRNA sequencing (scRNA-seq) data, we unveiled collagen as the top signaling molecule derived from FBs influencing CM and Vas_EC development. Subsequently, we used a Pdgfra-CreER controlled diphtheria toxin A (DTA) system to ablate the FBs at different stages. We found that the ablation of FBs disrupted myocardium and vasculature development and led to embryonic heart defects. Using scRNA-seq, we further profiled the ablated hearts and identified molecular defects in their ventricular CMs and Vas_EC compared to control hearts. Moreover, we identified a reduction of collagen in the ablated hearts and predicted collagen as the major signaling pathway regulating the differentially expressed genes in the ablated ventricular CMs. Finally, we performed both short-term and long-term fibroblast ablation at the neonatal stage. We found that short-term

ablation caused a reduction in collagen and Vas_EC density, while long-term ablation may induce compensatory collagen expression without causing heart function reduction. In summary, our study has identified the function of fibroblasts in regulating myocardium and vasculature development and implicated an important role for the collagen pathway in this process.

Introduction

Fibroblast is one of the most abundant cardiac cell types and plays important roles in normal heart function and pathological heart remodeling at the adult stage^{1,2}. The fibroblast composition in the heart has been analyzed with multiple approaches such as flow cytometry, with a focus on the left ventricle^{3,4}. However, the anatomical location of fibroblasts in the heart along the developmental progression is still unclear. The heart in mice starts to develop into a four-chambered structure at E9.5. The atrial and ventricular chambers are connected by the atrioventricular canal (AVC), a transient structure that develops into the septum and valve cells^{5,6}. Although the main fibroblast population does not develop until E13.5⁷, fibroblast-like cells with the expression of *Col1a1* and other fibroblast genes start to develop in the AVC at E9.5⁶.

CM and Vas_EC are the other two major cell types in the heart. Vas_EC develop at about the same time as FB, while CM develop before the appearance of FB at the chambers. FB has been shown to couple with CM through gap junctions and promote their maturation in human three-dimensional micro-tissue consisting of FB, CM, and EC⁸. In mice, epicardium-derived cells, mostly fibroblasts, have been reported to serve as guidepost cells by secreting the chemokine *Slit2*, guiding Vas_EC to the correct anatomical locations and promoting their maturation⁹. However, the lack of direct assessment of FB function *in vivo* prevents a comprehensive understanding of its role in regulating CM and Vas_EC development.

To study the function of a cell type rather than a gene, toxin genes such as NTR and DTA have been engineered into conditional ablation systems to eliminate specific groups of cells^{10,11}. The DTA system has been used in mice to ablate cardiac progenitor cells and cardiomyocytes at early developmental stages¹². Recently, a group of proliferating fibroblasts with the expression of *Postn* was identified in neonatal hearts. Further elimination of this cell population using the DTA ablation system showed its importance in promoting cardiomyocyte maturation¹³. Additionally, the DTA system has been used to ablate fibroblasts at the adult stage to assess their function in adverse cardiac remodeling^{14,15}. However, the function of the primary fibroblast population during embryonic and neonatal stages remains unclear.

One of the major functions of FB is to synthesize and secrete extracellular matrix (ECM) proteins, which influence the behavior of other cell lineages^{2,16}. The ECM is heterogeneous and composed of various types, such as collagen, fibronectin, elastin, and laminin, each with different subtypes. For example, collagen has 28 genes in vertebrates¹⁷. The ECM plays important roles in heart development and regeneration. For instance, *hyaluronan and proteoglycan link protein 1 (Hapln1)* is critical in regulating myocardial compaction and heart regeneration in zebrafish¹⁸. ECM has also been found to regulate mouse CM proliferation through $\beta 1$ -integrin signaling in a cell co-culture assay¹⁹. Moreover, collagen has been shown to be essential for heart regeneration in adult zebrafish and mice²⁰. However, the heterogeneity and function of ECM genes in mouse heart development remain unclear.

In this study, we identified the anatomical patterns of cardiac FB, CM, and Vas_EC during embryonic development. We further examined the ligand-receptor interactions among these cell types and identified collagen as the top signaling pathway. Additionally, we investigated fibroblast function at different embryonic stages using a cell ablation system and analyzed the resulting

molecular defects with scRNA-seq. Finally, we performed short-term and long-term fibroblast ablation at the neonatal stage and found that fibroblasts are dispensable for normal heart function development. In summary, this study provides a comprehensive understanding of the role of cardiac fibroblasts in myocardium and coronary vasculature development during embryonic and neonatal stages.

Results

Cellular analysis of cardiac fibroblast anatomical location at different stages

To understand the spatial distribution of cardiac FBs along the developmental progression, we stained the fibroblast marker gene *Col1a1* on heart sections at different stages from E11.5 to P3. We observed a strong expression of *Col1a1* in epicardial cells at all analyzed stages (Fig. 1A). Inside the heart, we found *Col1a1*-positive cells enriched at the boundary of the atrial and ventricular regions in E11.5 and 12.5 hearts (Fig. 1A, S1), indicating their presence in the atrioventricular canal (AVC). By E13.5 and 14.5, the AVC cells had developed into valve structures, and the valve interstitial cells highly expressed *Col1a1* (Fig. 1A, S1). In the ventricles, a small proportion of *Col1a1*-positive cells were observed adjacent to the epicardium. In contrast, few *Col1a1* signal was found in the atrium at this stage (Fig. 1A). At E15.5 and 16.5, both ventricles were filled with *Col1a1*-positive cells, but the middle of the ventricular septum remained devoid of such cells (Fig. 1A, S1). FBs were also found in the atrium, but the signal at the tips of both atrial chambers was sparser (Fig. 1A, S1). The distribution of FBs at E17.5 and E18.5 was highly similar to that observed at E16.5 (Fig. 1A, S1). Furthermore, we analyzed the *Col1a1* signal in P0, P2 and P3 hearts and found dense *Col1a1* expression at the valves and in all four chambers (Fig. 1A, S1). In summary, this analysis revealed the spatial location of cardiac FBs along the developmental stages. Specifically, we found that the FBs developed first in the AVC and valve, then at the ventricular wall and atrial, and last at the ventricular septum (Fig. 1B).

Next, to analyze the spatial relationship between FB and other cardiac cell types, such as CM and EC, we performed RNA staining for their marker genes *Col1a1*, *Actn2*, and *Cdh5*. Consistent with the previously described pattern, we observed that FB were mainly enriched in valves and began to develop at the edges of ventricular chambers at E13.5. We also found that FB were intermingled with CM populations and spatially adjacent to ECs (Fig. S2A). Next, we stained the three lineage markers on E17.5 and P3 heart sections. We observed consistent results as E13.5 (Fig. 1C, S2B). Given that FB starts emerging at E13.5, we primarily quantified the contacts at E17.5 and P3. The quantification results showed that each FB was in contact with approximately one Vas_EC and four CMs at both stages. (Fig. 1C, D). These results suggest that cardiac FB are physically close to CMs and ECs, likely facilitating signal communications that impact their development.

Identification of heterogeneity in cardiac fibroblasts through the analysis of scRNA-seq data

Through the analysis of scRNA-seq data at 18 stages of developing hearts from CD1 mice, we investigated the expression of *Col1a1* and observed high expression in epicardial cells and fibroblasts (Fig. S3A). This is consistent with the findings from RNA staining (Fig. 1A). Subsequently, we reanalyzed the fibroblasts and identified four distinct populations comprising 13 clusters (Fig. S3D). Gene expression analysis revealed that all cells in these populations exhibited high expression of *Col1a1* (Fig. S3A). Moreover, through differential gene expression analysis between the populations, we discovered that one group of cells highly expressed *Hapln1* (Fig. S3B, D, E), a marker gene associated with valves. This indicates that these cells are valve interstitial cells. We also found another group of cells expressing *Cdh5* and *Tie1* (Fig. S3B, D, E), genes associated with the endothelial cell lineage. This suggests that these cells may represent

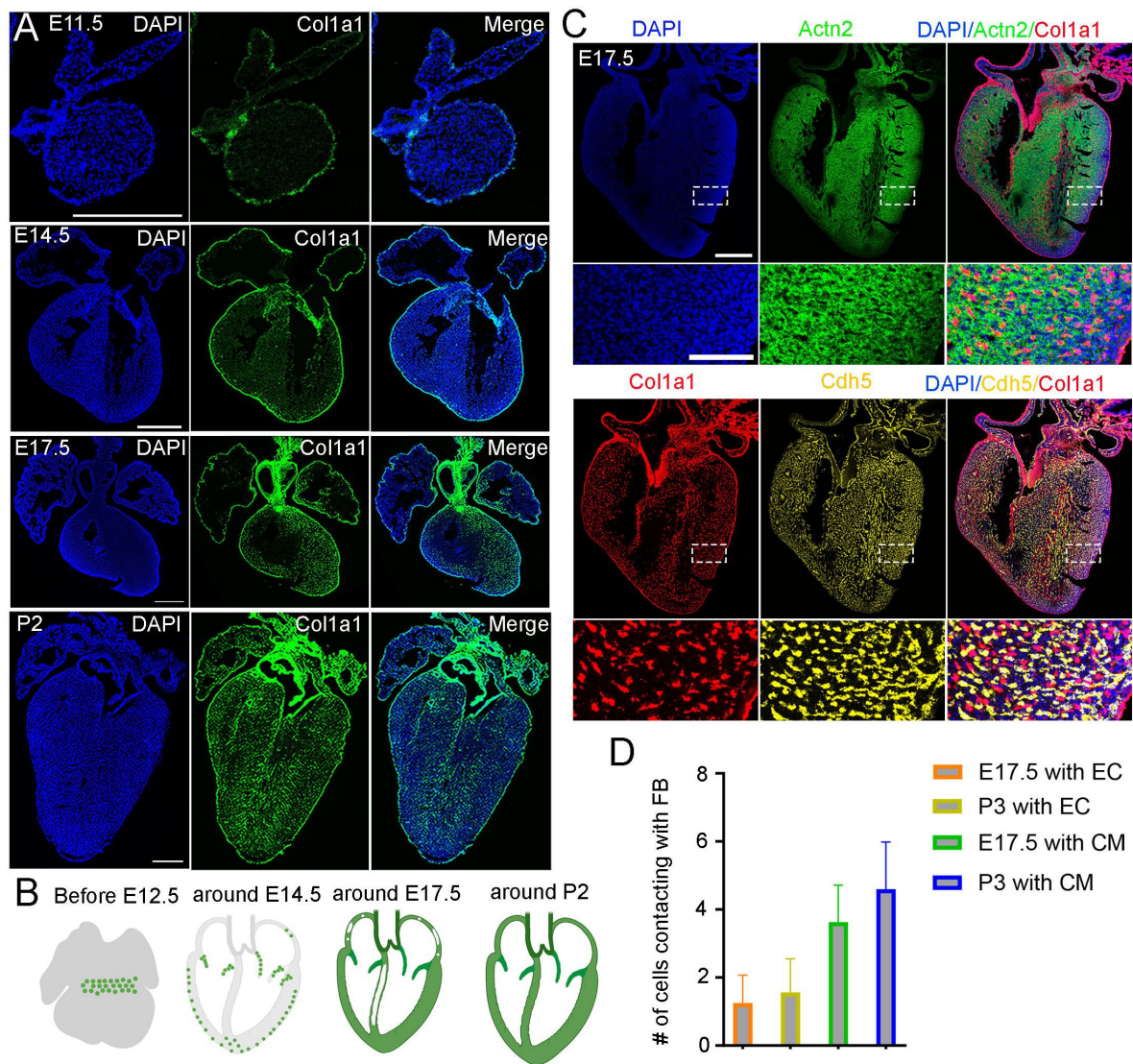


Fig 1.

Anatomical location of cardiac fibroblasts and their spatial relationship with cardiomyocytes and endothelial cells at different stages.

(A) RNA staining of *Col1a1* revealed the spatial pattern of cardiac fibroblasts at different developmental stages. Scale bar=500μm. (B) The development of cardiac fibroblasts can be grouped into four phases. (C) RNA staining analysis of *Col1a1*, *Actn2*, and *Cd31* revealed the spatial proximity of FB, CM, and EC in E17.5 hearts. (D) Quantification of the number of CMs and ECs that contact with each FB (n=100). Scale bar=500μm and 100μm in the whole heart sections and enlarged sections, respectively.

fibroblasts derived from EDCs. Additionally, we identified a third group of fibroblasts expressing *Sox10* and *Phox2b* (Fig. S3B, D, E [↗](#)), genes associated with the neural crest cell lineage. This suggests that these cells may originate from neural crest cells. Finally, the rest of the cells formed a main population that likely represents epicardial cell-derived fibroblasts in the chambers, as they expressed *Tcf21* and other marker genes (Fig. S3D, E [↗](#)). Interestingly, further analysis of the cell cycle phases in these fibroblasts revealed that each population consisted of cells in all three phases (Fig. S3C [↗](#)), indicating active proliferation during the early stages.

Next, to understand the anatomical location of these fibroblast populations, we performed RNA in situ hybridizations on postnatal heart sections. Firstly, we stained *Hapln1* together with *Dcn*, another Pan-fibroblast marker gene. We found that *Hapln1* was highly expressed in valves, and most of its signal overlapped with *Dcn* (Fig. S3Fi [↗](#)), suggesting its expression in valve interstitial cells (fibroblast-like cells in valves). We also stained *Cdh5* with *Col1a1* and observed a small cluster of cells with the expression of both genes, mostly located at the boundary of large vessels and ventricular chambers (Fig. 2Fii [↗](#)). Finally, we co-stained *Sox10* and *Col1a1* expression (Fig. S3Fiii [↗](#)). As previously reported²¹[↗](#), we identified double positive cells in the outflow tract-derived large vessels (Fig. S3Fiii_1 [↗](#)). However, interestingly, we also observed their presence in coronary vessels (Fig. S3Fiii_2 [↗](#)), suggesting that these cells may have a broader function than previously thought. In summary, through the analysis of scRNA-seq data and in situ hybridizations, we identified different populations of cardiac FBs, which differentially expressed lineage genes likely from their precursor cells.

Molecular analysis of the signaling communications between fibroblast and the other cardiac cell types

To understand the interactions between FB and other cardiac cell types, we analyzed scRNA-seq data from developing CD1 hearts. By quantifying the number and strength of these interactions, we found that each cell type, except blood cells, had a similar number of interactions. However, in terms of interaction strength, FB exhibited the highest values compared to those of other cell types (Fig. 2A [↗](#), Supplementary table S2 [↗](#)). Next, we analyzed the signaling pathways secreted from FB to ventricular cardiomyocytes (Ven_CM) and Vas_EC. Interestingly, we found that the collagen pathway was the predominant pathway in both FB-VenCM and FB-VasEC interactions (Fig. 2B [↗](#)). In addition to collagen, we identified other signaling molecules such as laminin, Ptn, MK, Fn1, and Thbs, which are mostly extracellular matrix (ECM) proteins, as being secreted by FB to function in Ven_CM and Vas_EC (Fig. 2B [↗](#)). Within the collagen pathway, we further analyzed the detailed ligand-receptor interactions and found more interaction pairs in FB-VasEC than in FB-VenCM. Some interactions were shared between the two cell types, but most were distinct (Fig. 2C [↗](#)). Consistently, all the shared collagen types have been previously studied and found to play a crucial role in heart development, primarily by regulating AVC and vasculature development²²⁻²⁶[↗](#).

To better understand the role of ECM genes in FB_VasEC and FB_VenCM interactions, we analyzed their expressions (GO number, 0031012) in the chamber-derived fibroblasts (main_fb) using the CD1 and C57BL/6 scRNA-seq datasets from 18 stages¹⁰[↗](#) (Fig. S4A-F [↗](#)). Interestingly, we identified five groups of ECM genes that displayed stage or chamber specific expression in both datasets (Fig. S5 [↗](#)-7 [↗](#), Supplementary table S3 [↗](#), 4 [↗](#)). Specifically, we found one group of genes (G1) that was highly expressed at stages before E14.5. This gene group includes signaling molecules such as *Slit2* and *Wnt5a*, and collagen genes such as *Col13a1* and *Col26a1* (Fig. S5 [↗](#)-7 [↗](#)). We found the second group of genes (G2) were mainly expressed in the LV and RV from E14.5 onwards with a significant increase at the neonatal stage. This gene group includes collagen genes such as *Col5a1*, *Col6a1*, and *Col6a2*, and metalloproteinase genes such as *Adamts13* and *Adamts14*. Additionally, we identified the third group of genes were highly expressed in the LA and RA at most stages, which includes *Col12a1*, *Adamts1*, and *Adamts4*. Interestingly, *Fn1* also exists in this group, suggesting that it has an important function in atrial fibroblast development. Furthermore, we

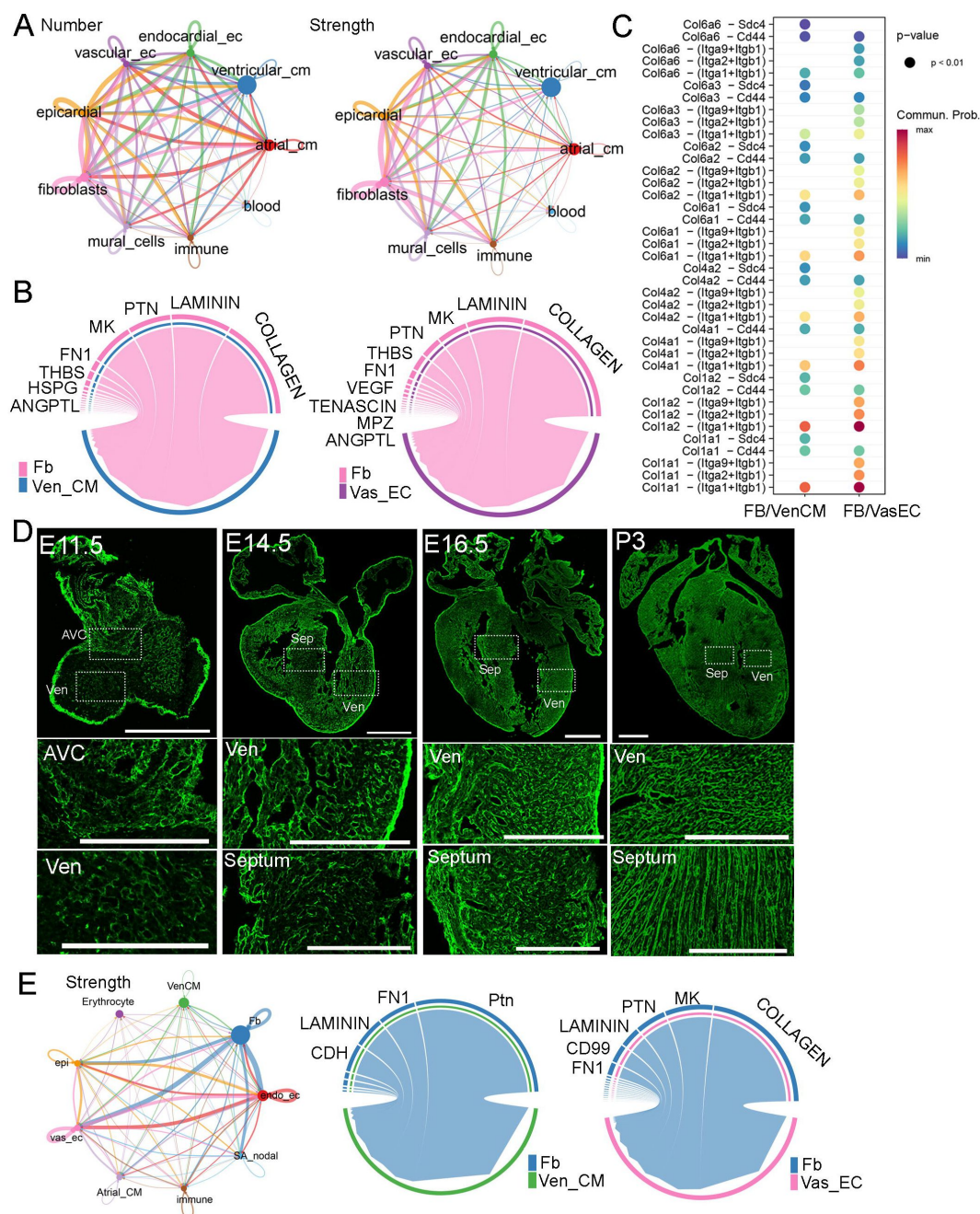


Fig 2.

Molecular analysis of fibroblast communications with other cardiac cell types in developing hearts.

(A) The number and strength of interactions among different cardiac cell types identified from scRNA-seq data. (B) The top signaling pathways derived from FB that impact Ven_CM and Vas_EC development. (C) The detailed ligand-receptor interactions in the collagen pathway between FB and Ven_CM or Vas_EC. (D) Collagen was visualized using CHP staining in mouse hearts at E11.5, E14.5, E16.5, and P3. Scale bar=500µm and 150µm in the whole heart sections and enlarged sections, respectively. (E) The signaling interactions and top pathways between FB and Ven_CM or Vas_EC in primary human hearts.

found the fourth group of genes were preferentially expressed in the other three chambers than the RA and were mainly expressed at late neonatal stages after P5 in certain chambers. This group of genes includes *Col4a4*, *Col4a5*, *Col4a6*, and *Ntn4*. Finally, we identified the fifth group of genes displayed universal expression in all four chambers at most stages. This gene group includes *Col1a1*, *Col1a2*, and *Postn*, which are the ubiquitously expressed extracellular matrix genes (**Fig. S5** [4](#)–[7](#)). In summary, we identified five groups of ECM genes displaying stage- and chamber-specific expression patterns in FB and observed that each gene group includes different collagen pathway members. Additionally, we analyzed the ligand-receptor interactions between FB and Ven_CM or Vas_EC at each stage. We found collagen pathway interactions were also enriched at most analyzed stages (**Fig. S8** [4](#), **Supplementary table S5** [4](#), [6](#)). These results together suggested the importance of collagen pathway in mediating the FB function in heart development.

Furthermore, we studied collagen deposition dynamics in developing hearts. We used collagen hybridizing peptide (CHP) to stain the total collagen after denaturing it via an antigen retrieval process. Consistent with *Col1a1* expression, we observed strong collagen signals in the epicardium at all analyzed stages (**Fig. 2D** [4](#)). Interestingly, at E11.5, we observed collagen in both AVC and ventricular chambers. Given that fibroblasts in the chambers have not developed yet at this stage, the observed collagen is likely derived from other cell types, such as endocardial endothelial cells. At E14.5, we observed a strong collagen signal in the ventricular free walls, while the signal in the ventricular septum and atrium was relatively weaker. At E16.5, the collagen signal became brighter and denser throughout the heart, including the ventricular free wall, septum, and atrium. This distribution was slightly broader compared to the fibroblast distribution at the same stage (**Fig. 1A** [4](#)). By P3, the collagen signal had further increased and formed stripe patterns (**Fig. 2D** [4](#)). In summary, we found that the progression of collagen accumulation largely coincided with the development of FBs.

Lastly, we analyzed the signaling interactions among human cardiac cells using a human embryonic heart scRNA-seq dataset²⁷ [4](#). Consistent with the mouse analysis results, we identified strong interactions between FB and other cardiac cell types (**Fig. 2E** [4](#)). Additionally, we found collagen to be the most enriched pathway in FB-VasEC interactions. However, interestingly, we did not observe the collagen pathway among the top enriched pathways in FB-VenCM interactions. This observation was confirmed in two other human embryonic heart scRNA-seq datasets²⁸ [4](#),²⁹ [4](#). These results suggest a partially similar mechanism underlying fibroblast regulation of heart development in both mice and humans.

Functional analysis of fibroblasts in embryonic heart development

To identify the right transgenic mice labeling the developing cardiac fibroblasts, we analyzed the expression of two well-known fibroblast marker genes, *Postn* and *Pdgfra*, in the CD1 scRNA-seq data. We observed that *Postn* was highly expressed in fibroblasts and mural cells. Among fibroblasts, *Postn* was expressed in all subpopulations (**Fig. S9Ai** [4](#)). On the other hand, *Pdgfra* was specifically expressed in fibroblasts, although its expression was relatively lower than *Postn*, especially in postnatal staged atrial cells (**Fig. S9Aii** [4](#)). To evaluate their efficiency in labeling fibroblasts, we bred *Postn*-CreER mice with *Rosa*-mTmG reporter mice and administered tamoxifen (200µg/g) to pregnant mice at E13.5 (**Fig. S9Bi** [4](#)). We collected the hearts at E17.5 and observed only a small proportion of GFP-positive cells in the valves and septum (**Fig. S9Bii** [4](#)), suggesting a low labeling efficiency from the *Postn*-CreER mice. In contrast, we conducted the same experiments using *Pdgfra*-CreER mice and observed a significant proportion of GFP-positive cells in both the valves and heart chambers (**Fig 3A** [4](#)). These results demonstrate the efficient labeling capability of the *Pdgfra*-CreER; *Rosa*-mTmG mice.

To study the function of fibroblasts in heart development, we utilized a genetically encoded DTA system to ablate them, which involves the encoding of a toxin gene that induces apoptosis and allows for the ablation of target cells. Initially, we crossed floxed *Rosa*-DTA mice with *Pdgfra*-CreER mice and administered tamoxifen to pregnant female mice at E10.5. After 3 days, we harvested the

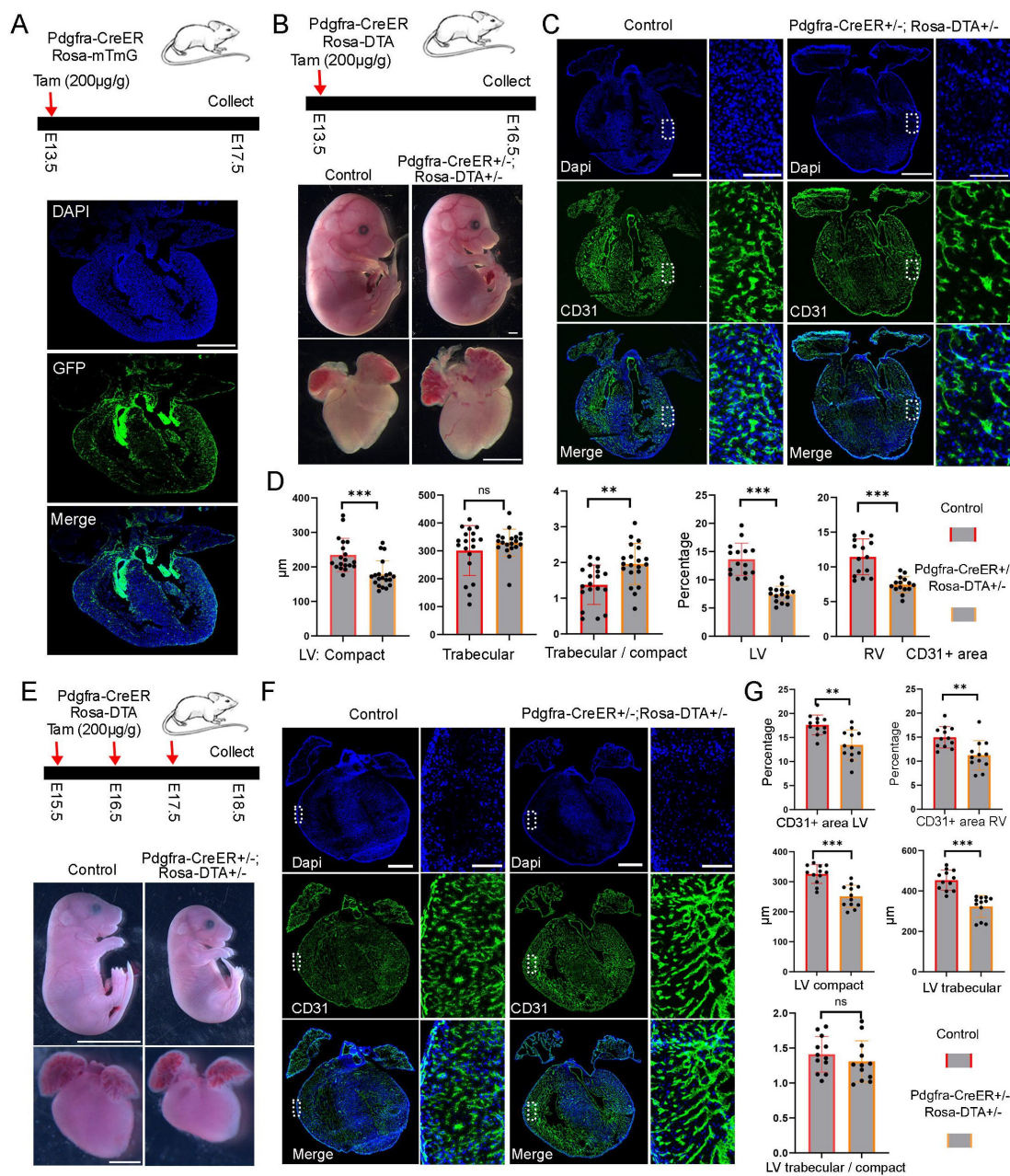


Fig 3.

Functional analysis of FB in embryonic heart development.

(A) Diagram of the lineage tracing experiments and the tracing results. Scale bar = 500µm. (B) The ablated embryos treated with tamoxifen at E13.5 were smaller than control embryos, but the ablated hearts were not obviously different from the control hearts. Scale bar=1mm. (C) Representative CD31 staining images in control and ablated hearts. The zoomed-in images showed a reduction of CD31 positive area in the ablated hearts compared to controls. (D) Quantification of the thickness of compact and trabecular myocardium (19 sections from 7 ablated hearts and 20 sections from 7 control hearts), and CD31 positive areas in control and ablated hearts (15 sections from 5 ablated and 15 sections from 5 control hearts). The littermate control mice include both double-negative and *Pdgfra-CreER*^{+/-} mice. (E) The ablated mice and hearts treated with tamoxifen at E15.5 were smaller than controls. (F) CD31 staining analysis of control and ablated hearts. (G) Quantification of the compact and trabecular myocardium thickness, and CD31 positive areas in control and ablated hearts (12 sections from 4 ablated hearts and 12 sections from 4 control hearts.). The littermate control mice include both double-negative and *Pdgfra-CreER*^{+/-} mice. Scale bar =500µm in the whole heart sections and 100 µm in the zoomed in images. * represents $p < 0.05$; ** represents $p < 0.01$.

embryos and observed that all the embryos with ablated fibroblasts (*Pdgfra-CreER*^{+/−}; *Rosa-DTA*^{+/−}) had perished, while the control embryos remained unaffected. This outcome indicates an essential role for fibroblasts in early-stage embryo development.

Subsequently, we administered tamoxifen to the female mice from the breeding pair at E13.5 and collected the embryonic hearts at E16.5 (**Fig 3B**). Our findings revealed that the ablated embryos exhibited smaller sizes compared to the control embryos, although the size of the ablated hearts did not significantly differ from that of the control hearts (**Fig 3B**). Additionally, we conducted TUNEL analysis to assess cell death in both control and ablated hearts and observed a significantly higher number of TUNEL-positive dots in the valve and chamber regions of the ablated hearts compared to the control hearts, confirming the successful ablation of fibroblasts in the experimental hearts (**Fig. S10A, B**). Furthermore, we examined the anatomical structure of the hearts but did not identify obvious defects in the ablated groups compared to control groups. However, we observed a reduction in endothelial cell density in the ablated hearts, as indicated by the CD31 staining signal (**Fig 3C**). Additionally, we measured the thickness of the compact and trabecular myocardium (**Fig S10C**). Our observations revealed that, compared to the control hearts, the ablated hearts exhibited a thinner left ventricular (LV) compact myocardium, while the thickness of the LV trabecular myocardium remained similar.

Furthermore, we noted an elevated ratio of LV trabecular to compact myocardium in the ablated hearts compared to the control hearts (**Fig 3D**). However, no similar differences were observed in the right ventricle (RV). Finally, we examined cell proliferation post-ablation using pHH3 staining. We observed a reduction in pHH3-positive cells in the ventricular region, but not in the atrial region, of the ablated hearts compared to the control hearts (**Fig. S10B**). These results suggest that cell proliferation in ventricle was impaired following fibroblast ablation.

Next, we performed ablation at late embryonic stages by treating the pregnant mice with tamoxifen at E15.5 and harvested the embryos at E18.5 (**Fig. S12A**). We did not observe obvious differences in the size of the embryos and hearts between the ablated group and the control group (**Fig. S12B**). However, the CD31 staining of heart sections revealed large holes in the ablated atrium but not in the ablated ventricles or control hearts (**Fig. S12C**). Additionally, quantification of the compact and trabecular myocardium thickness in the LV and RV revealed a thinner LV compact myocardium (**Fig. S12D**). We also calculated the ratio of trabecular to compact myocardium thickness in the LV and RV and found that the ratio was increased in the RV but not in the LV of the ablated hearts compared to control hearts (**Fig. S12E**). These results indicated that both the LV and RV of the ablated hearts have subtle defects. Moreover, we analyzed cell proliferation by staining pHH3 and found no significant differences between the ablated and control ventricles (**Fig. S12F**).

Given that the animals did not show obvious morphological changes after one dose of tamoxifen treatment, we increased the tamoxifen treatment to three consecutive days, ranging from E15.5 to E17.5 (**Fig 3E**). We harvested the embryos at E18.5 and found that both the ablated embryos and their hearts were notably smaller in size compared to the controls (**Fig 3E**). Next, to confirm the ablation efficiency, we analyzed *Col1a1* expression with RNA staining and found that the ablated hearts lost almost all of their *Col1a1* signal in the four chambers (**Fig. S12G**), indicating a high ablation efficiency with three doses of tamoxifen treatments. We further analyzed their valve structures and observed that the mitral valves in the ablated hearts appeared shorter than their counterparts in the control hearts. Specifically, we found that the mitral valve leaflets in control hearts were already connected to the papillary muscle, whereas the same valve leaflets in ablated hearts were not (**Fig. S11A**). Moreover, by conducting a staining analysis of CD31 in the control and ablated heart sections, we discovered that the ablated hearts exhibited a significantly lower endothelial cell density (**Fig 3F**, G). Further quantification of the LV compact and trabecular myocardium thickness showed a significant reduction in the ablated hearts compared to control hearts (**Fig 3G**). However, interestingly, the ratio of trabecular to compact myocardium

thickness in the LV did not differ between the control and ablated hearts, indicating a proportional reduction in both compact and trabecular myocardium thickness in the ablated hearts. Finally, we found that the ablated ventricles had less pHH3 signal than the control ventricles (**Fig. S12F**), indicating impaired cell proliferation in the ventricles following fibroblast ablation. These results together indicated that fibroblast plays an important role in embryonic heart development at all the analyzed stages.

Molecular analysis of the defects in the ablated embryonic hearts

Next, we performed scRNA-seq to analyze the defects in the ablated embryonic hearts. We utilized MULTI-seq to pool the control and ablated hearts from E16.5 and E18.5 (**Fig. 4A**). After filtering out low-quality cells based on our QC standards, we grouped the single cells from two experimental replicates using unsupervised methods and found a high degree of overlap (**Fig. 4B**, **S13A**). We then identified cell types and cell cycle phases within the single cells. In total, we identified eight cell types: Atrial_CM, Ven_CM, Vas_EC, Endo_EC, Mural_cell, FB, Epicardial cell, and Immune cell, along with three cell cycle phases (G1, S, G2M) in each cell type (**Fig. 4C**). Cells from both control and ablated conditions at E16.5 and E18.5 were present in each cell type. We further quantified the percentages of each cell type across conditions and found that the percentage of FB was significantly reduced in the ablated hearts at both stages, indicating effective ablation. Interestingly, we also observed a reduction in mural cells in the ablated hearts at both stages (**Fig. 4D**).

Next, we selected FB for further study of its cellular heterogeneity. By analyzing the expression of cardiac lineage genes, cell cycle phases, and sample sources, we identified four FB subpopulations (**Fig. 4E**). Specifically, we identified a group of valve fibroblasts (valve_fb) expressing valve mesenchymal cell genes such as *Hapln1*, a group of neural crest-derived fibroblasts (Neural_crest_fb) expressing *Sox10*, and a group of chamber-derived main fibroblasts (main_fb) expressing *Wt1* and *Tcf21* (**Fig. S13B**). All these FB groups contained cells from both control and ablated samples. In contrast, we identified another group of cells exclusively derived from the ablated samples, which expressed cell death-related genes (**Fig. S13C**). We named this group dying fibroblasts (dying_fb).

In addition to FB, we analyzed Ven_CM and found that cells from control and ablated hearts were highly overlapped on the UMAP plot (**Fig. 4F**). We quantified the proportions of cells at each cell cycle phase and found that the ablated samples had slightly higher percentages of cells in the G1 phase compared to control samples (**Fig. 4G**, **S13D**). We then performed differential gene expression analysis of control and ablated samples at E16.5 and identified a set of differentially expressed genes (**Fig. 4H**, **Supplementary table S7**, **8**). Gene ontology analysis of the upregulated genes in the ablated CMs showed enrichment in various pathways, including heart development, sarcomere organization, heart tube morphogenesis, and cell proliferation. In contrast, the downregulated genes were enriched in pathways related to chaperone-dependent protein folding, mitochondrial membrane permeability, fatty acid import, and mitochondrial ATP transmembrane transport (**Fig. 4I**). These results indicated that ablated Ven_CMs upregulated genes associated with heart development and cell proliferation while downregulating genes related to cell maturation, including those involved in mitochondrial function and cell metabolism.

Furthermore, we selected Vas_EC for further analysis. We did not observe distinct populations between control and ablated samples on the UMAP plot (**Fig. 4J**, **S13E**). However, through further differential expression analysis, we identified a group of genes with upregulated expression in the ablated samples (**Fig. 4K**, **Supplementary table S9**). Gene ontology analysis of these genes revealed that they were primarily involved in immune response pathways, including type I and II interferon production, interleukin-27-mediated pathways, NLRP3 inflammasome assembly, and cellular responses to exogenous dsRNA (**Fig. 4L**). These results suggest that Vas_EC exhibits a strong immune response following fibroblast ablation.

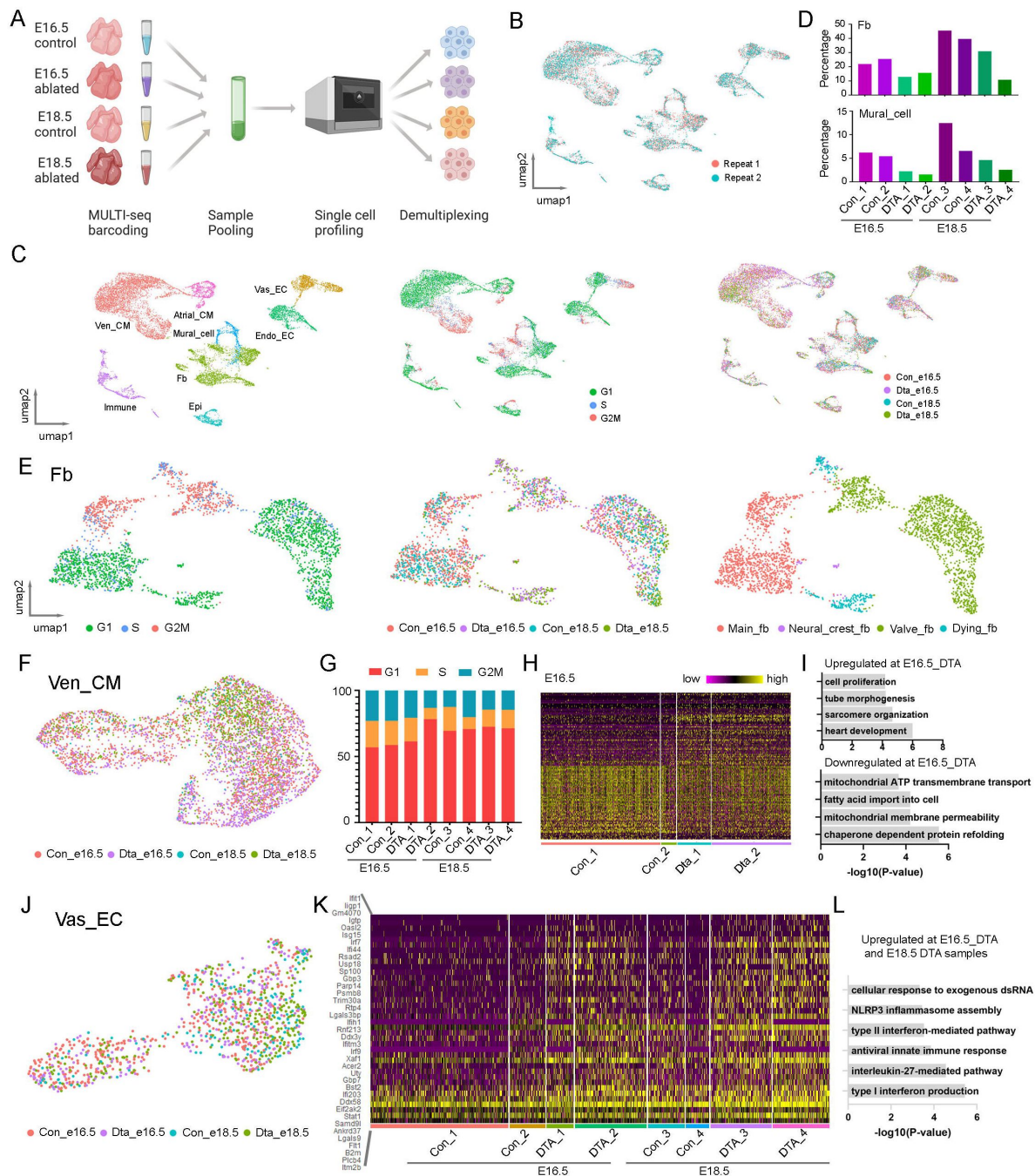


Fig 4.

Single cell analysis of the embryonic heart defects after fibroblast ablation.

(A) Diagram of the MULTI-seq experiments to profile control and ablated hearts at two developmental stages. (B) The scRNA-seq data from two replicates are highly consistent. (C) UMAP plots of scRNA-seq data labeled by cell type, cell cycle phase, and genotype. (D) Quantification of the FB and mural cell percentages at each conditions. (E) Detailed analysis of the fibroblast population revealed four groups, including a dying fibroblast subpopulation. (F) UMAP plot of Ven_CMs from different conditions. (G) Quantification of the different cell cycle phased Ven_CMs under each condition. (H, I) Heatmap and pathway enrichment of genes that are differentially expressed in control and ablated Ven_CMs at E16.5. (J) UMAP plot of Vas_EC labeled by conditions. (K, L) Heatmap and pathway enrichment of genes that were upregulated in ablated Vas_EC compared to control Vas_EC. The littermate control mice at both stages were *Pdgfra-CreER*^{+/−}.

The signaling pathway defects in the ablated embryonic hearts

Next, we studied the signaling defects in the ablated hearts. Through the analysis of signaling pathways between main_fb and Vas_EC or Ven_CM, we found that the collagen pathway consistently ranked at the top in both control and ablated samples (**Fig. 5A**). We then stained for collagen levels using collagen hybridizing peptide and observed that collagen deposition was dramatically reduced in both the septum and ventricular wall of the ablated hearts compared to the control hearts (**Fig. 5B**).

Moreover, by analyzing the scRNA-seq data, we predicted the signaling ligands that may regulate the differentially expressed genes in control and ablated Ven_CM. Interestingly, the collagen pathway, which has four different collagen ligands on the list, emerged as a major regulator of these genes (**Fig. 5C**). In addition to collagens, we also identified other ligands, such as TGF-beta, IGF, and BMP, which were derived from FB and regulate the expression of differentially expressed genes in control and ablated Ven_CMs (**Fig. 5C**). We also predicted the receptors in Ven_CM for each ligand (**Fig. S14A**). Furthermore, we predicted the ligands in FB that regulate the differentially expressed genes in Vas_EC and identified several signaling molecules. These included the DTA pathway-related signal Hbegf, the inflammation-associated signal Il33, Wnt ligands (Wnt11, Wnt4, and Wnt5b), and collagen Col4a6 (**Fig. 5D**, **S14B**).

Finally, we predicted the signals from the dying_fb cells. Compared to normal main_fb and valve_like_fb, the dying_fb downregulated certain signaling pathways, such as the IGF pathway, EphA pathway, Tenascin pathway, and collagen pathway (**Fig. 5E**, **S15A**). Interestingly, they also upregulated several signaling pathways, including the EGF pathway, BMP pathway, VEGF pathway, and FGF pathway (**Fig. 5E**, **S15B**). Understanding the role of these altered signaling pathways from dying_fb in contributing to heart defects will be an interesting area for future research. Additionally, collagen and other ECM proteins are essential structural components. Their roles as structural proteins in heart development are critical and contribute to the heart phenotype observed in the ablated hearts.

Functional analysis of fibroblasts at neonatal stages

Next, we assessed the fibroblast function at neonatal stage. We treated the mice with tamoxifen at P1 and harvested them at P4 (**Fig. S16A**). We did not observe any obvious differences between the ablated and control hearts (**Fig. S16B**), consistent with a recent study that used the same strategy to ablate fibroblasts during a similar time period³⁰. However, CD31 staining of heart sections revealed defects in the atrium, which had a thinner and segmented atrial wall (**Fig. S16C**). We further quantified the compact and trabecular myocardium in both the LV and RV but did not observe differences between control and ablated hearts (**Fig. S16D**). We also did not find differences in the ratio of trabecular to compact myocardium thickness between control and ablated hearts in the LV and RV (**Fig. S16E**). Finally, we analyzed cell proliferation by staining pHH3 and did not observe differences between the two conditions (**Fig. S16F**). Next, we increased the tamoxifen treatments to three times, ranging from P1 to P3, and collected hearts at P4 (**Fig 6A**). The heart morphology and size did not reveal obvious differences between control and ablated groups (**Fig 6B**). The CD31 staining revealed a reduction in endothelial cell density in the ablated hearts when compared to the control hearts (**Fig 6C**). These reductions in both LV and RV were further confirmed through quantifications (**Fig 6D**). Finally, we quantified the chamber thickness and cell proliferation. Similar to hearts treated with one dose of tamoxifen, we observed no apparent defects and cell proliferation changes in ablated hearts with three days of tamoxifen treatments (**Fig 6D**, **S16G**). Furthermore, we stained the P4 hearts for collagen depositions. We observed that the ablated hearts had a sparser collagen signal and lost most of the stripe pattern in the septum compared to the control hearts. However, a significant amount of

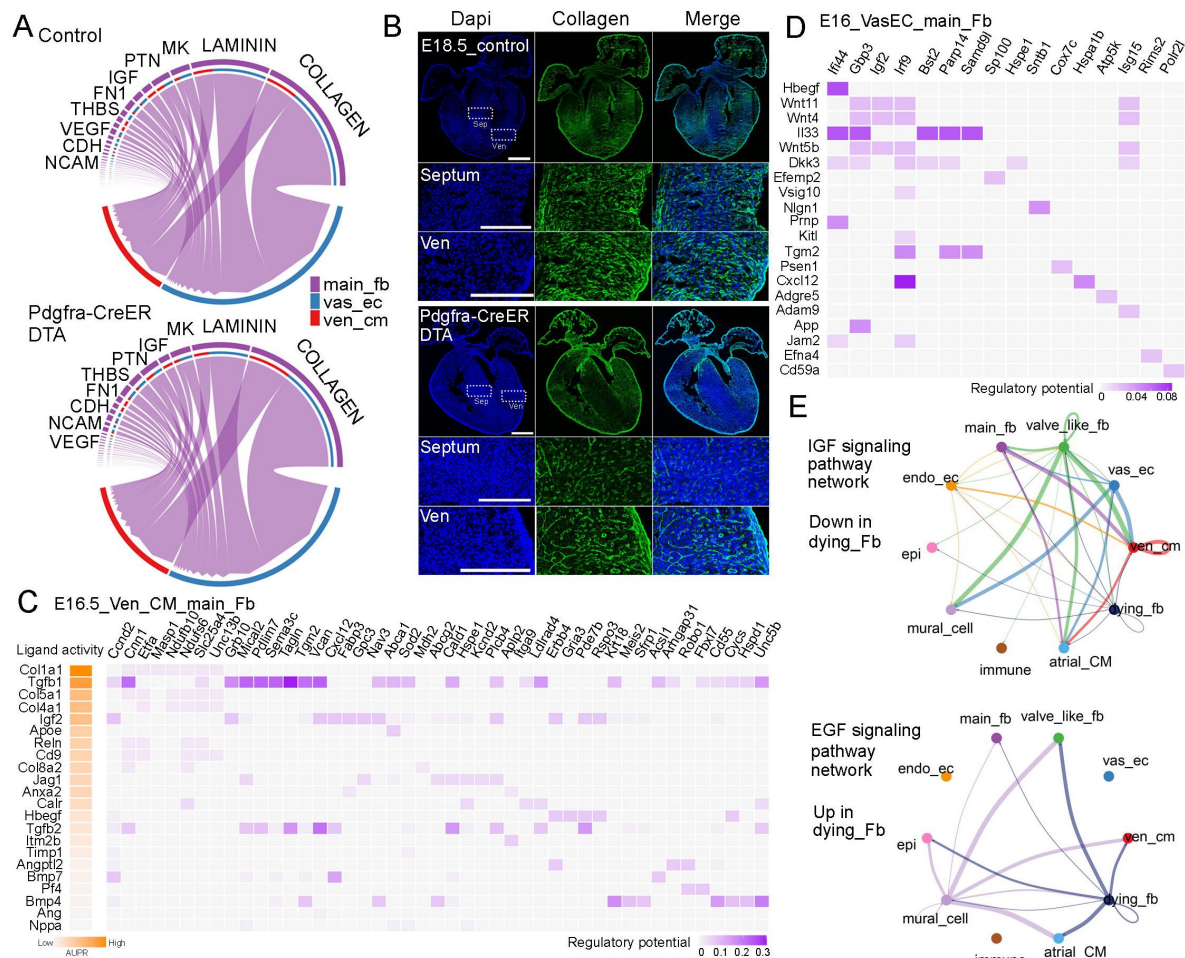


Fig 5.

Identification of signaling defects in ablated hearts.

(A) The top signaling pathways between FB and Ven_CM or Vas_EC in control and ablated hearts. (B) CHP staining revealed a significant reduction in collagen deposition in ablated hearts compared to control hearts. (C, D) Regulatory analysis predicted the ligands that regulate the genes differentially expressed in control and ablated Ven_CMs and Vas_EC. (E) Representative signaling pathways that were downregulated or upregulated in the dying FBs.

stripe-patterned collagen was still observed in the ventricular free wall, although the overall signal was sparser (**Fig. 6E**). These collagens were likely derived from fibroblasts before ablation.

Lastly, we conducted a long-term analysis by administering tamoxifen to the mice from P3 to P5 and harvesting them at P17 or P18 (**Fig. 6F**). We observed that approximately two-thirds of the ablated mice had perished before P17 (**Fig. 6G**), and the surviving mice were significantly smaller than the control mice. Quantification revealed reduced heart rates and body weights, but a similar heart-to-body ratio in the ablated mice compared to controls (**Fig. 6I**). A detailed examination of the major tissues showed reduced organ sizes in the ablated mice, including smaller hearts, lungs, brains, kidneys, and livers (**Fig. 6F**, **S18A**). Additionally, we assessed mouse heart function using echocardiography before sacrificing them. Interestingly, we found that heart function, including ejection fraction (EF) and fractional shortening (FS), was not reduced in the ablated mice compared to controls (**Fig. 6H, I**). Furthermore, we evaluated collagen levels in the ablated hearts and did not observe a clear reduction or pattern change (**Fig. S17**). This finding contrasts with our observations at P4 (**Fig. 6E**) and suggests potential collagen expression compensation at later stages. In contrast, we found that the lungs of the ablated mice exhibited reduced collagen levels and larger empty spaces compared to the controls (**Fig. S18B**). These findings suggest that fibroblasts play a crucial role in the growth of multiple tissues during the neonatal phase. Additionally, it will be interesting to explore the mechanisms underlying the preservation of heart function when fibroblasts are eliminated in future studies.

Discussion

In this study, we investigated the cellular and molecular interactions between FB and CM or Vas_EC, identifying collagen as the primary signaling molecule. We also analyzed the role of FB in embryonic heart development using a cell ablation system. Additionally, we employed scRNA-seq to identify the molecular defects in CM and Vas_EC within the ablated hearts. Finally, we examined the function of FB in neonatal heart development through both short-term and long-term ablation experiments. Altogether, our study represents the first comprehensive analysis of fibroblast function in heart development. The insights we gained from the study will not only enhance our understanding of the role of this major cardiac cell type in heart development but also shed light on its function in adult cardiac remodeling, as well as other tissue development and regeneration.

Through RNA staining of the three lineage genes, we found that each fibroblast directly interacts with approximately one Vas_EC and four CMs. Given that the quantification was conducted on 2D tissue sections, the number of contacts may be underestimated. Considering that Vas_EC and FB emerged around the same time, it will be important to understand in the future how their connections are established at the single-cell level and what these connections mean for the development of each lineage. Two studies in mice and zebrafish have shown that epicardial cells or epicardium-derived cells regulate Vas_EC development through the expression of *Hapln1* or *Slit2*^{8,19}. It would be interesting to test whether collagen or other fibroblast-derived ECM proteins have similar roles. Additionally, *in vitro* culture experiments have demonstrated that fibroblasts interact with CMs through gap junction proteins or integrins to regulate their proliferation^{8,19}. Future studies should aim to clarify how these interactions are established *in vivo* and how they evolve during developmental progression.

Collagen, as the major ECM family, consists of 28 members¹⁷. In addition to collagen, there are many other types of ECM genes. Our scRNA-seq data revealed the temporal and spatial expression patterns of ECM genes in cardiac FBs. Future research should investigate their differential functions. For example, we found that most ECM genes in group 4 upregulate their expression by postnatal day 7, when mouse hearts begin to lose their regenerative capacity, suggesting a potential role in heart regeneration. This warrants further investigation. Additionally, while we

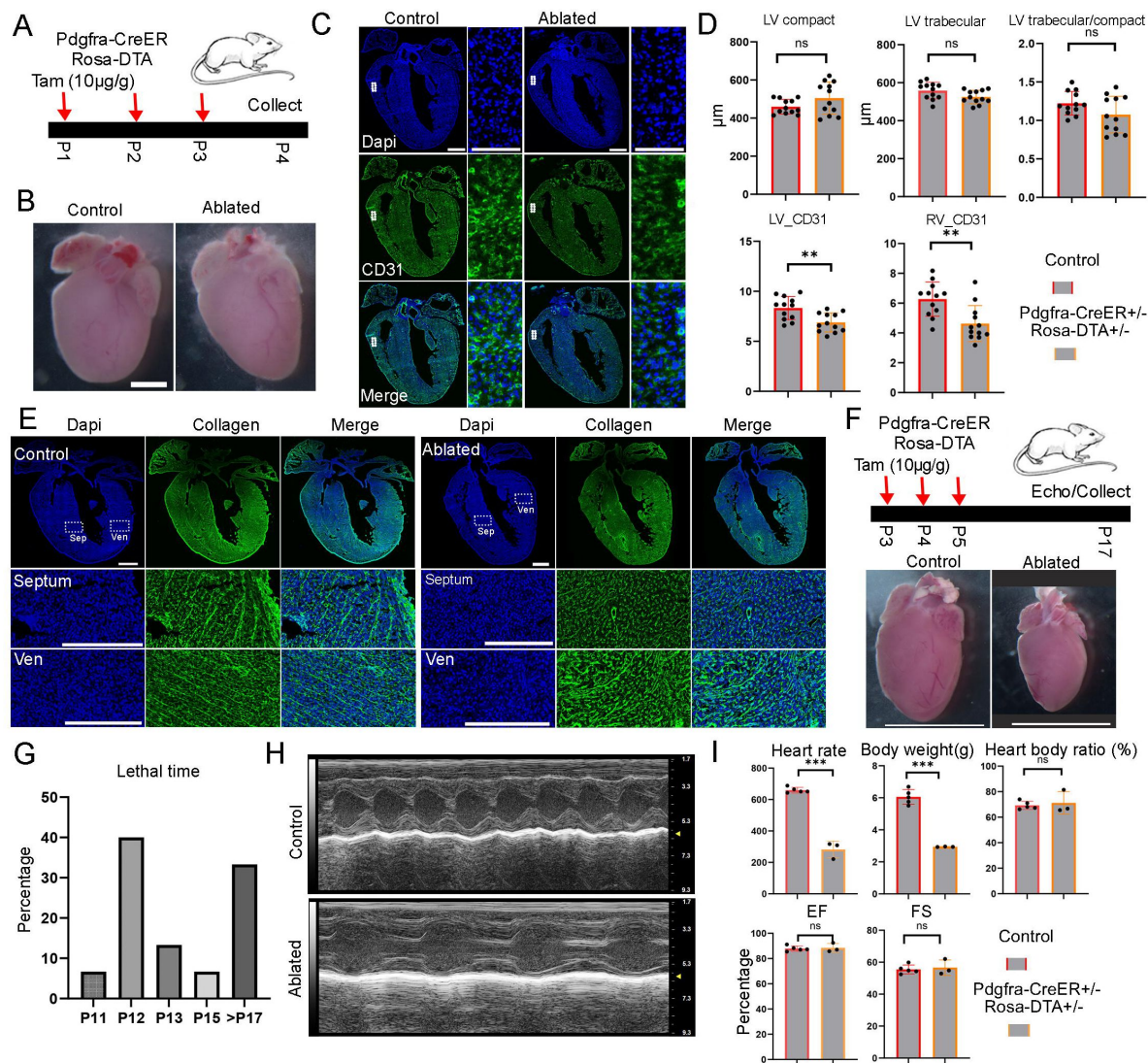


Fig 6.

Short term and long term functional analysis of fibroblasts at neonatal stage.

(A) Diagram of the experiment to ablate fibroblasts with three doses of tamoxifen treatment from P1 to P3. Scale bar=1cm. (B) No obvious size difference was observed between control and ablated hearts. (C) Representative images of control and ablated hearts stained with CD31. (D) Quantification of the compact and trabecular myocardium thickness and CD31 positive areas in ventricular at control and ablated hearts (12 sections from 4 ablated hearts and 12 sections from 4 control hearts.). (E) Collagen accumulation in control and ablated hearts at P4. Collagen was visualized using CHP staining. The littermate control mice were Pdgfra-CreER^{+/−}. Scale bar=500µm and 150µm in the whole heart sections and enlarged sections, respectively. (F) Tamoxifen was given from P3 to P5 to ablate the fibroblasts and hearts were collected at P17 for analysis. The littermate control mice were Pdgfra-CreER^{+/−}. (G) The percentage of mice died at different days. (H) Representative echo images of the control and ablated hearts. (I) Quantification of the heart rate, body weight, and heart function in control and ablated mice at P17. * represents $p<0.05$; ** represents $p<0.01$; *** represents $p<0.001$.

have analyzed the collagen pathway as a whole in this study, our scRNA-seq analysis indicated that collagen genes in different groups exhibit distinct expression patterns. It will be essential to explore their unique roles in heart development in future studies. Furthermore, we found that collagen is the primary signaling molecule in FB_CM and FB_VasEC pairs in mice; however, we only observed it in FB_VasEC pair in humans. This discrepancy may be due to the loss of collagen receptors in human CMs, which merits further exploration.

The DTA-based cell ablation system has been used to ablate cardiac progenitors and cardiomyocytes before. It revealed that a robust regeneration response occurred after cell ablation at early stages¹². However, in our study, based on the pHH3 staining, we did not observe an increase in cell proliferation after cell ablation. This could be due to multiple possibilities. First, the *Pdgfra*-CreER; Rosa-DTA-based ablation may be too severe to initiate the response to cell loss. Considering that ablation at E13.5 and ablation with three doses of tamoxifen at E15.5 both led to smaller embryos, the ablation of a large number of cells may compromise the compensation mechanisms. However, it is difficult to explain the embryos at E15.5 with one dose of tamoxifen treatment, which did not display obvious defects after ablation and have no increase in cell proliferation. Second, unlike cardiomyocytes, fibroblasts may have a limited compensatory response to their own ablation, although they can respond to CM loss at both neonatal and adult stages.

ScRNA-seq results revealed the upregulation of inflammation-related pathways in Vas_EC in the ablated hearts. It will be important to investigate whether these pathways are initiated by the ablation process or the loss of FBs in the future. Given that these pathways were primarily upregulated in Vas_EC but not in CM, this suggests a cell type-specific response. Studying the function of these signals in vascular development will be essential. Moreover, pathway analysis revealed both upregulation and downregulation of signaling pathways associated with dying fibroblasts. Further studies on these pathways are important, as they could offer valuable insights into the cell death process during normal heart development and heart injury and repair.

Fibroblast ablation at the neonatal stage led to two-thirds of the mice experiencing a lethal outcome, while the remaining mice developed a dwarf phenotype. Functional analysis of the mouse hearts showed preserved cardiac function post-ablation (**Fig. 6H, I**), suggesting that the lethal phenotype may be due to defects in other tissues, such as the lungs, which exhibited disrupted structures. Additionally, we conducted functional measurements at P18 under anesthesia in a small cohort of mice and found that the ablated mice exhibited a trend of improved heart function (**Fig. S19**). This finding indicates that fibroblasts may play a role in regulating heart function during the neonatal stage, aligning with observations in adult mice. In the adult stage, cardiac function was found to be superior in residential fibroblast-ablated mice compared to control mice following myocardial infarction or angiotensin II/phenylephrine (AngII/PE) treatments¹⁴.

Methods

Experimental methods

Mouse strains

The animal experiments have been approved by the University of Pittsburgh Institutional Animal Care and Use Committee (IACUC). CD1 male and female mice were purchased from Charles River Laboratories and bred in our laboratory to generate embryos and neonatal pups at specific stages

for RNA staining experiments. The transgenic mice, including Rosa26-mTmG (Strain #:007676)³¹, Pdgfra-CreERT2 (Strain #:032770)³², Postn-CreER (Strain #:029645)³³, and ROSA26-eGFP-DTA (Strain #:032087)¹⁰, were ordered from the Jackson Laboratory.

Tamoxifen treatment and mouse dissection

The male and female mice of specific strains were bred together. The default dosage of 200µg of tamoxifen per gram of body weight (200µg/g) was given to pregnant mice through oral gavage, and neonatal mice were given 10µg/g of tamoxifen by direct injection into their stomach to induce Cre activity¹³. The pregnant mice and neonatal pups were euthanized using CO2 and decapitation-based methods, respectively. The mouse hearts were isolated following the standard procedure described previously and were directly embedded in OCT without fixation for RNA staining or fixed at 4% paraformaldehyde for other experiments, such as immunofluorescence staining.

Proximity ligation in situ hybridization (PLISH)

PLISH was performed following a published protocol³⁴. Specifically, the embryonic or postnatal mouse hearts were embedded in OCT (Sakura, 4583) without fixation. After sectioning at a thickness of 10µm, the tissue sections were treated with post-fix medium (RNase-free PBS with 3.7% formaldehyde and 0.1% DEPC) followed by 0.1mg/ml pepsin treatment (RNase-free H2O with 0.1mg/ml pepsin and 0.1M HCl). After dehydration, the sections were sealed with hybridization chambers (Invitrogen, S24732) and hybridized with H probes (Supplementary table S1) in Hybridization Buffer (1 M NaTCA, 5 mM EDTA, 50 mM Tris pH 7.4, 0.2 mg/mL Heparin). Next, after being treated with circularization reaction and rolling cycle amplification, the samples were hybridized with detection probes conjugated with Cy3 or Cy5 fluorophore. Finally, the samples were stained with DAPI (Invitrogen, D1306), mounted with fluoromount-g (SouthernBiotech, OB100-01), and imaged under confocal microscopy (Leica TSC SP8).

RNAscope Multiplex Fluorescent V2 Assay

The RNAscope Multiplex Fluorescent Reagent Kit v2 (Advanced Cell Diagnostics, 323270) was performed according to the manufacturer's manual. Briefly, the tissue sections were fixed in pre-chilled 4% PFA (Electron Microscopy Sciences, 15710) at 4°C for 1 hour. After progressive dehydration, the sections were sequentially treated with hydrogen peroxide for 10 minutes and protease IV for 15 minutes (for embryonic) or 20 minutes (for postnatal) at room temperature (RT). After that, the samples were hybridized to gene-specific Z probes for 2 hours at 40°C using the HybEZ II Hybridization system (ACD, 321721). Following further signal amplification, the hybridization signals were detected with TSA Vivid Fluorophores. The samples were then stained with DAPI and mounted with ProLong Gold Antifade Mountant (Invitrogen, P36930), and imaged under confocal microscopy (Leica TSC SP8). The RNAscope probes used in the study are Mm-Col1a1-C2 (319371-C2), Mm-Actn2 (569061), and Mm-Cdh5-C3 (312531-C3).

Quantification of the interacting cells

Based on the RNA scope staining results, the direct contacts between fibroblasts and cardiomyocytes as well as endothelial cells was quantified. Given that the fibroblast starts to emerge at E13.5, we mainly quantified the contacts at E17.5 and P3. Approximately 100 fibroblasts from the left ventricular wall at each stage were selected, and the number of *Actn2*⁺ and *Cdh5*⁺ cells surrounding each fibroblast was counted respectively.

Immunofluorescence staining

The immunofluorescence staining was performed following a standard procedure. Briefly, mouse hearts were fixed in 4% PFA overnight before embedding in OCT. Afterwards, the samples were sectioned at 10µm and briefly washed in PBS to clean the OCT. The sections were then blocked for

1 hour in blocking buffer (10% goat serum, 1% BSA, 0.1% Tween 20) and incubated with primary antibodies in the primary antibody buffer (1% BSA in PBST) at 4°C overnight. On the second day, the samples were stained with fluorophore-conjugated secondary antibodies in blocking buffer for 1 hour at room temperature. Finally, the samples were stained with DAPI, mounted with fluoromount-g, and imaged with a confocal microscope. The primary antibodies used in the study include anti-CD31 (BD, # 550274), and anti-pHH3-488 (abcam, #ab197502). The secondary antibodies used are Goat Anti-Rabbit IgG-488 (ThermoFisher, # A-11008) and Goat Anti-Rabbit IgG-647 (ThermoFisher, #A21247).

Collagen staining

Fresh frozen sections from embryonic and postnatal mouse hearts were fixed in 4% PFA in 1X PBS for 15 minutes at room temperature (RT). After that, the sections were soaked in a sodium citrate-based solution (distilled water with 10mM sodium citrate and 0.5% Tween-20, HCl was added to adjust pH to 6.0) for antigen retrieval, which was performed in a steamer for 30 minutes. Subsequently, the sections were incubated with 20 μ M biotin-conjugated collagen hybridizing peptide (Advanced Biomatrix, 50-196-0307) in 1X PBS at 4°C overnight. The next day, streptavidin-cy5 (Invitrogen, SA1011) was used (1:500 in 1X PBS with 1% BSA) for 1 hour at RT. Finally, the samples were stained with DAPI, mounted with FLUOROMOUNT-G, and imaged under a confocal microscope.

TUNEL staining

TUNEL staining was performed following the manufacturer's protocol (Roche, 11684795910). Briefly, mouse hearts were embedded and sectioned as described in the immunofluorescence staining procedure. After fixation with 4% Paraformaldehyde in PBS for 20 minutes and three washes with PBS, the sections were incubated in permeabilization solution (0.1% Triton X-100, 0.1% sodium citrate) for 2 minutes on ice, followed by two PBS rinses. Subsequently, 60 μ l of TUNEL reaction mixture was added to each section, and they were incubated in a humidified atmosphere for 1 hour at 37°C in the dark. Finally, the sections were washed with PBS three times, stained with DAPI, mounted with fluoromount-g, and imaged with a confocal microscope.

Single cell mRNA-sequencing experiments

ScRNA-seq was performed following the MULTI-seq procedure previously reported^{35,36}. Briefly, tamoxifen-treated E16.5 and E18.5 Pdgfra-CreER; Rosa-DTA mouse embryos were harvested on the same day. Since the ablated mouse embryos at both stages were smaller than the control embryos, we selected one control and one ablated embryo from each stage for heart dissection. The genotype of the selected embryos was further validated using PCR. The hearts, with four chambers, were dissociated into single cells and labeled with MULTI-seq barcodes. The pooled libraries were then loaded onto the 10X Genomics Chromium iX and profiled using the single-cell 3' V3.1 kit. The generated libraries were sequenced on the Illumina NovaSeq X Plus platform. The experiments were repeated twice.

Echo imaging

Echocardiography was performed using a standard protocol³⁷. Briefly, heart function of awake mice at P17 was assessed using the Vevo 2100 micro imaging platform (FUJIFILM Visual Sonics Inc., Canada). The measurements were conducted at the level of papillary muscle in M mode. Heart rates, left ventricular end-diastolic diameter (LVEDD), and left ventricular end-systolic dimensions (LVESD) were calculated from the 2D short-axis view.

Statistical analysis

To quantify pHH3 signal, compact, and trabecular myocardium thickness, we used two to three sections from each heart and more than three hearts per genotype. The exact number of embryos profiled can be found in each figure legend. Statistical analyses were conducted using ImageJ and Prism 9 software, employing a two-tailed Student's t-test to compare groups. P-values below 0.05 were considered significant.

Data analysis

Data Availability

The newly generated scRNA-seq datasets have been deposited in the Gene Expression Omnibus (GEO) under the accession number GSE272048. Please use this secure token (sdcxiewmxzfytar) to review the data. The scRNA-seq datasets from CD1 and C57BL/6 mouse hearts were generated in a previous study and can be downloaded from the GEO database using the accession number GSE193346³⁶. The human scRNA-seq data were obtained from GEO or the European Genome-Phenome Archive (EGA) under the accessions GSE106118 and EGAS00001003996^{27–29} for CellChat analysis.

ScRNA-seq data analysis

ScRNA-seq data were mapped to the mouse genome mm10 using CellRanger and further demultiplexed with the R package deMULTiplex³⁵. Cell types in the newly generated scRNA-seq data were annotated using lineage genes previously published by our team³⁶. We performed quality control by removing cells with fewer than 200 genes and those with more than 40% mitochondrial content. Experimental batch integration was conducted using Harmony 1.2.0³⁸, followed by unsupervised clustering with Seurat V5³⁹. Next, we utilized the Cell Cycle Scoring function in Seurat to annotate the cell cycle phases for each individual cell. Finally, we applied the FindMarkers function in Seurat with default settings to identify genes that were differentially expressed in Ven_CM or Vas_EC under control and ablation conditions.

Extracellular matrix genes expression analysis

We downloaded the extracellular matrix genes from the Jackson Laboratory under the term "extracellular matrix" and gene ontology ID 0031012 (<https://www.informatics.jax.org/go/term/GO:0031012>; Download date: 2023-07-07). After downloading, we cleaned the gene list by removing duplicates and filtering out genes with null expression, resulting in a total of 440 genes. Next, we assessed the enrichment of each gene in the main population of fibroblasts at each stage and chamber using the R package AUCell⁴⁰, with heatmaps drawn with the R package ggplot2 (Fig 9, S9, S10).

Ligand-receptor interaction analysis

To identify ligand-receptor interactions across stages and zones for fibroblasts and cardiomyocytes, we analyzed relevant subsets of the CD1 data, including the main population of FB, ventricular CM, and Vas_EC using CellPhoneDB v3.34⁴¹. We used a p-value threshold of 0.2 and ran the analysis with 10 threads, while keeping the rest of the parameters at their default settings. The significant mean value of all interactive partners (log2) and enrichment p-values (-log10) obtained from the CellPhoneDB outputs were plotted as dotplots in R.

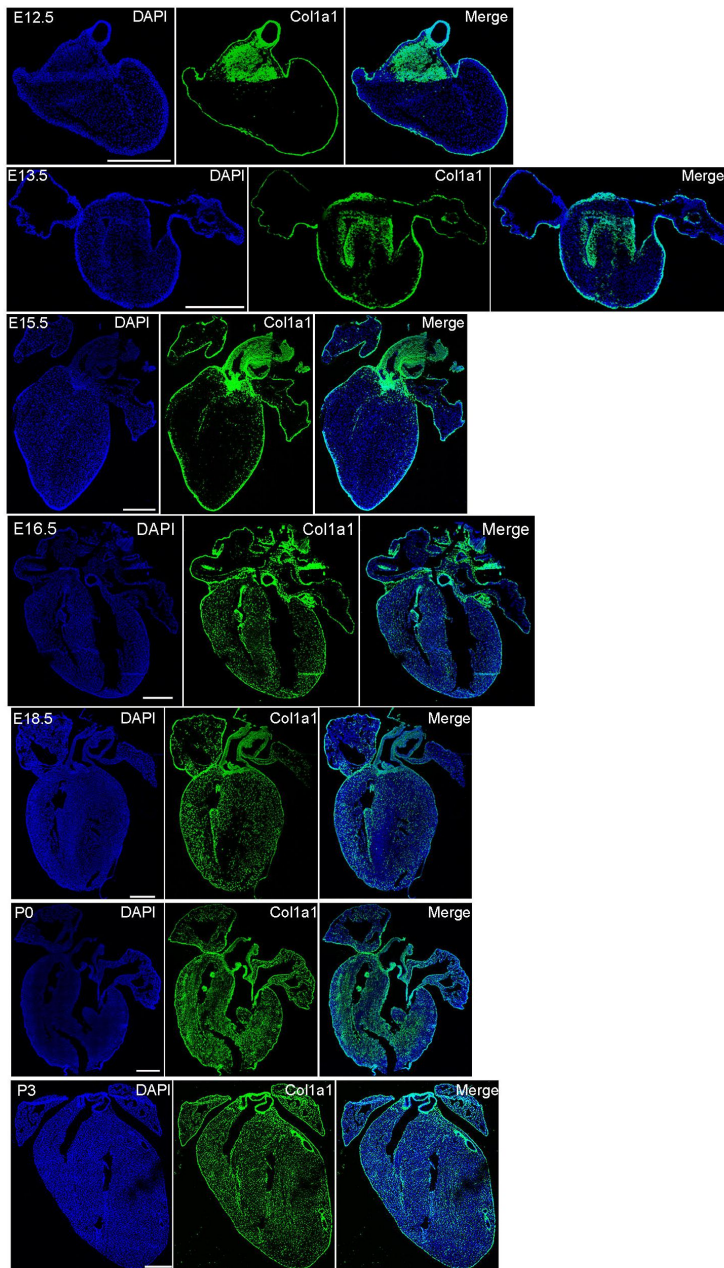
Cellchat analysis

The Seurat object containing mouse CD1 single-cell mRNA sequencing data was converted into a CellChat object in R. Default settings in CellChat 1.6.1⁴² were used to identify the number and strength of interactions among cell types. FB were designated as sender cells, while Ven_CM and Vas_EC were designated as receiver cells to identify signaling between these related cell types. The same default settings were applied to analyze signaling pathways among human cardiac cell types. For pathway analysis in control and DTA-ablated cardiac cells, their respective scRNA-seq objects were utilized.

NicheNet analysis

Nichenetr 2.1.5⁴³ was used for the analysis with default settings. First, we identified differentially expressed genes between control and ablated cells. Due to the varying number of recovered cells in each cell type under different conditions, we selected different staged samples for comparison and assessed their consistency across the remaining samples. For Ven_CMs, we used the E16.5_control_1 and E16.5_DTA_1 samples, while for Vas_EC, we included all four E16.5 samples. Next, we designated the control sample as the background condition and the ablated sample as the experimental condition, using main_fb as the sending cells and Ven_CM or Vas_EC as the receiving cells to predict their regulatory potential.

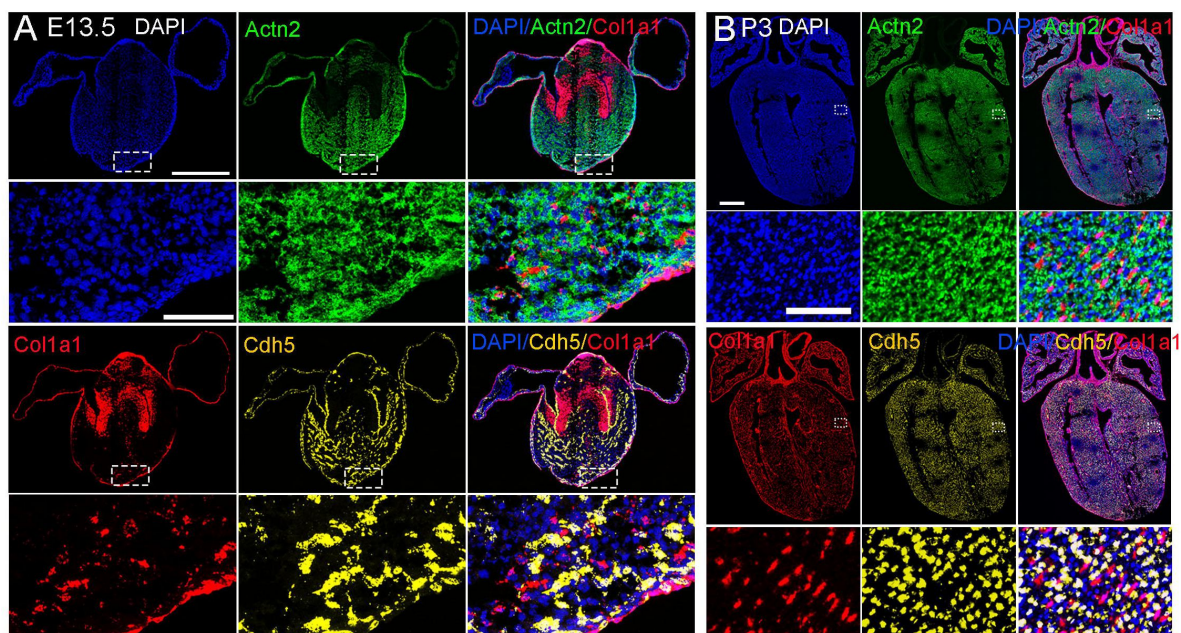
Supplementary figures



Supplementary Fig. S1

(A) Analysis of fibroblast anatomical pattern by staining Col1a1 at different stages.

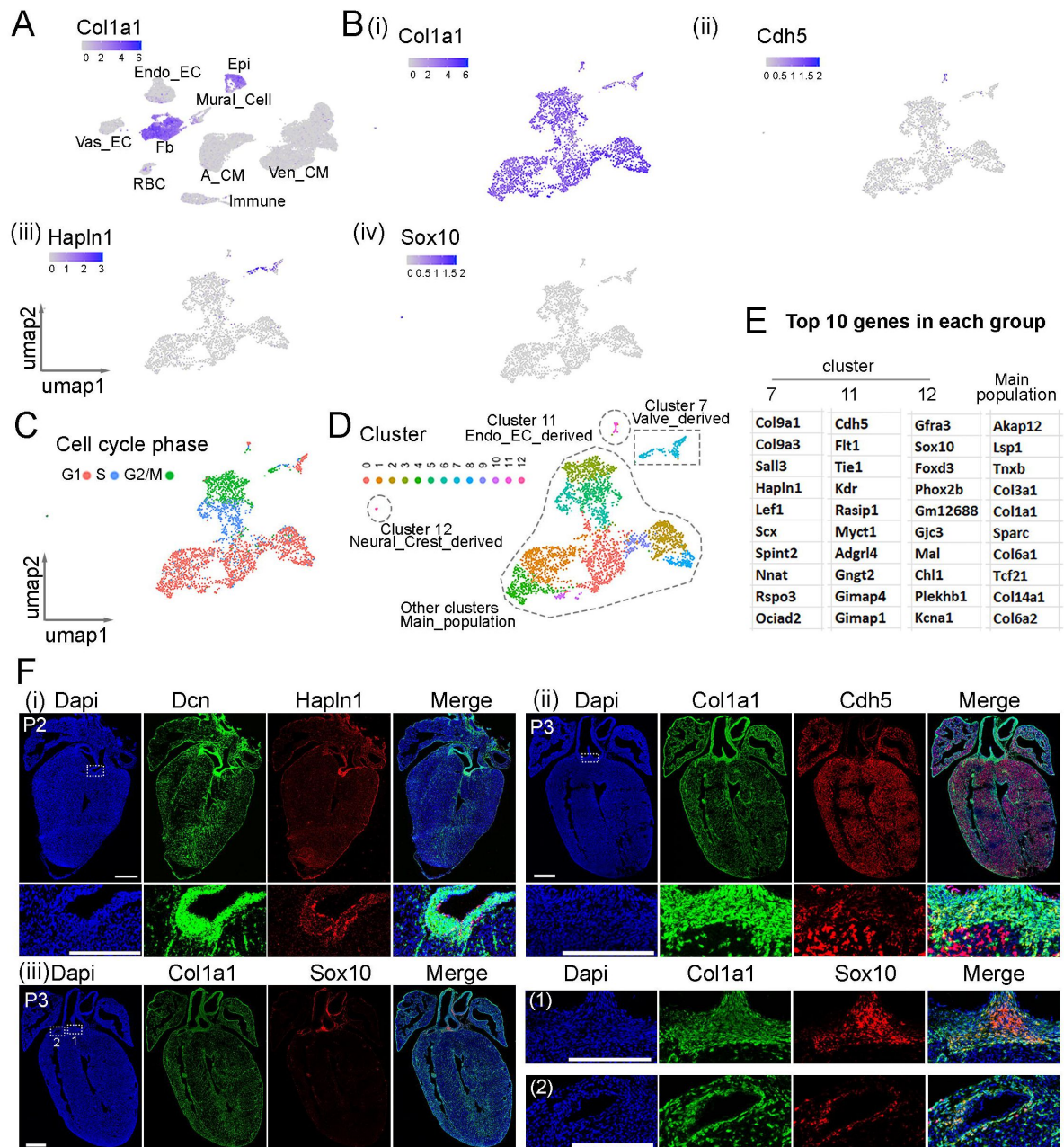
Scale bar=500μm.



Supplementary Fig. S2

>(A, B) RNA staining of *Col1a1*, *Actn2*, and *Cdh5* expression at E13.5 and P3 heart sections.

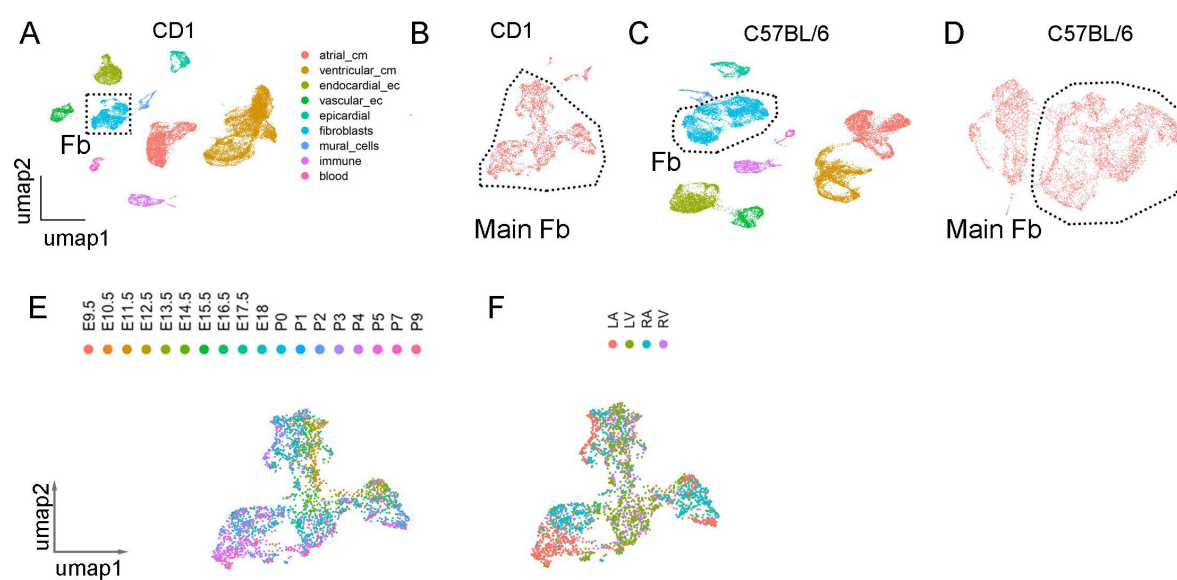
Scale bar=500µm and 100µm in the whole heart sections and enlarged sections, respectively.



Supplementary Fig. S3

scRNA-seq identified distinct fibroblast populations.

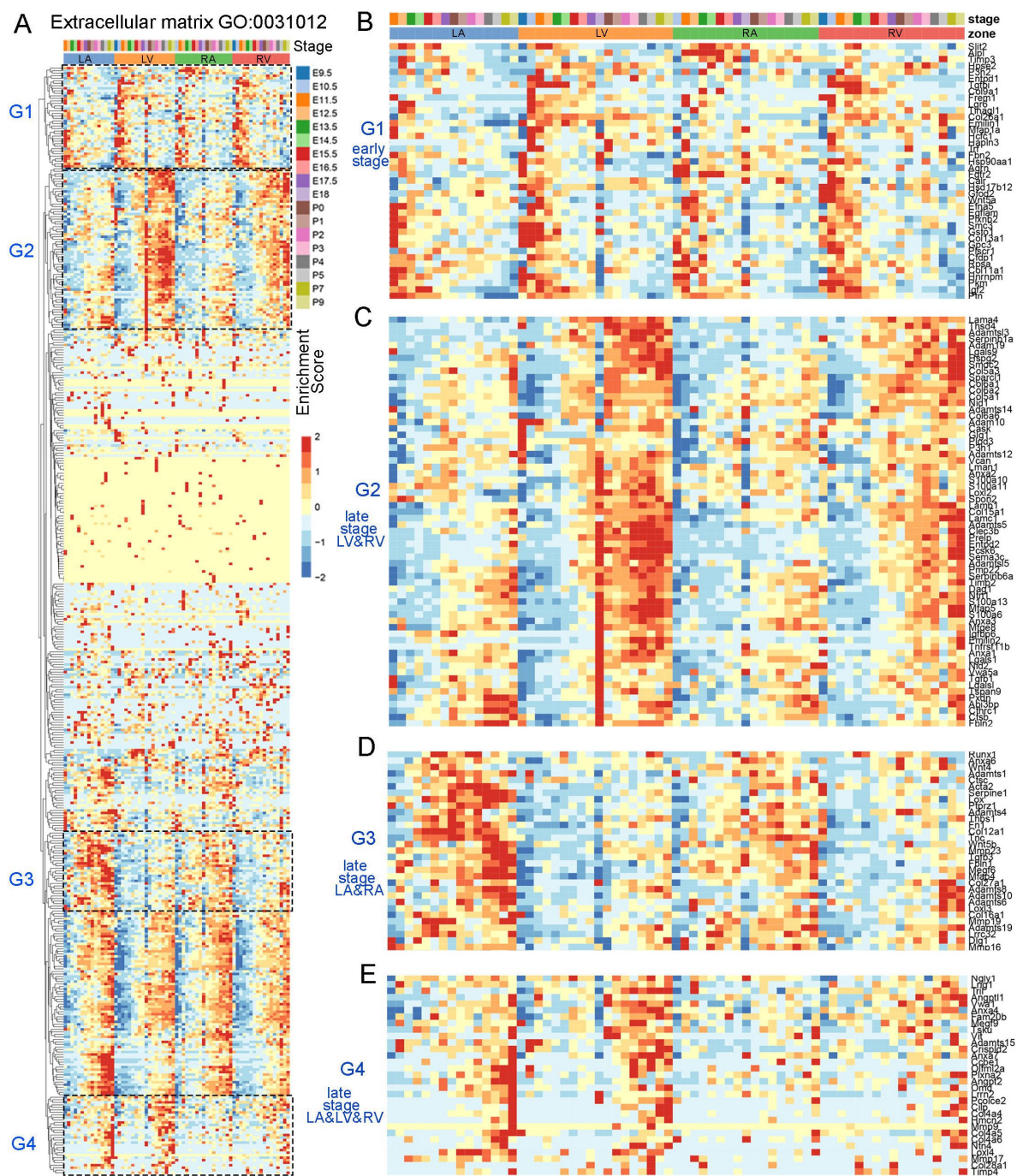
(A) UMAP plot of *Col1a1* expression in cardiac cells. (Bi-iv) UMAP plots of *Col1a1* and representative cluster-specific genes expression in cardiac fibroblasts. (C) UMAP plot of fibroblasts labeled by cell cycle phases. (D) Diagram of the four types of cardiac fibroblasts. (E) The top 10 genes expressed in each group of cardiac fibroblasts. (Fi-iii) Identification of the anatomical location of each fibroblast population through RNA staining. Scale bar=500µm and 250µm in the whole heart images and the images with enlarged areas, respectively.



Supplementary Fig. S4

Heterogeneity analysis of cardiac fibroblasts.

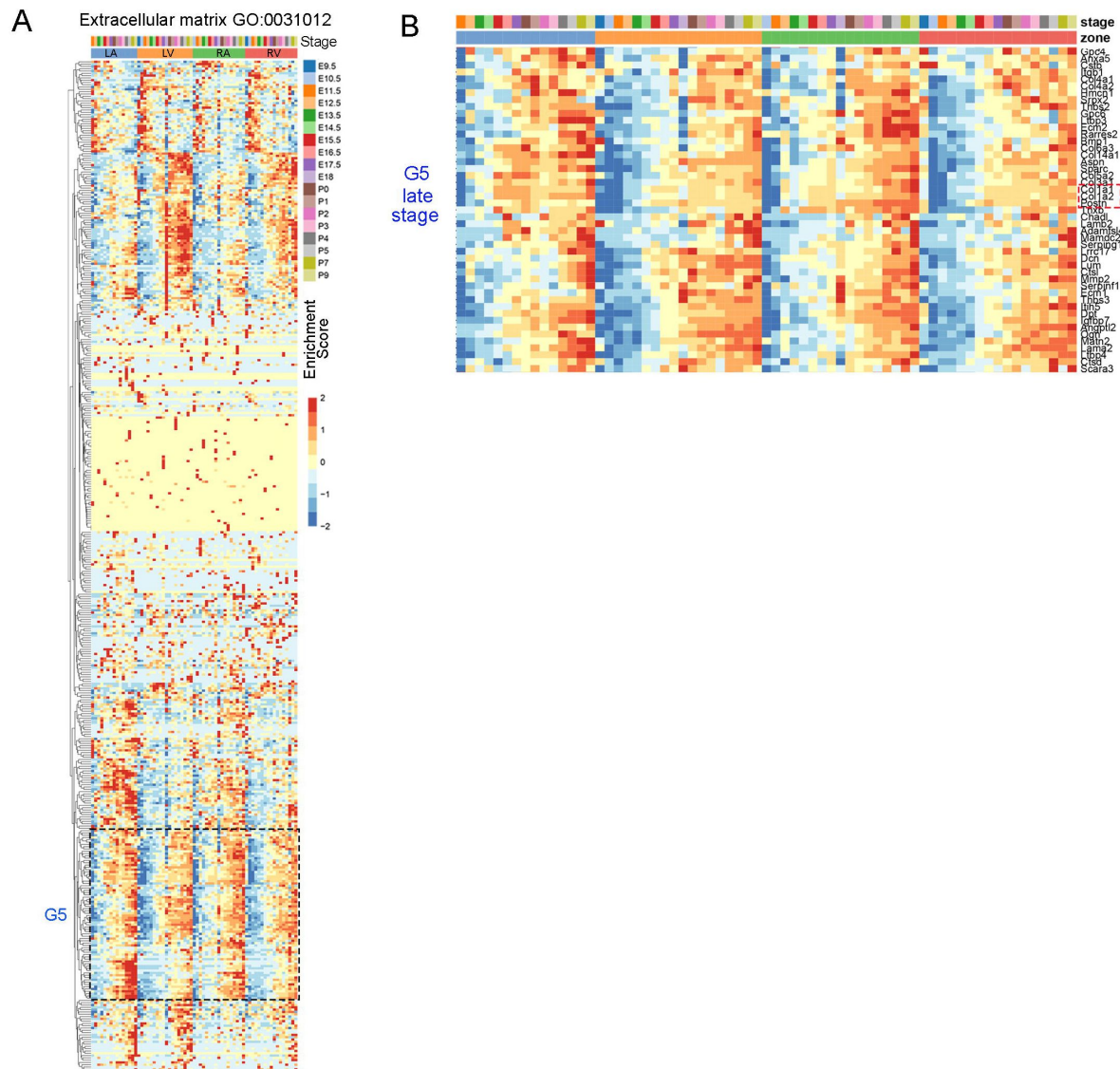
(A, B) The cell types and main_fib population in CD1 scRNA-seq dataset. (C, D) The cell types and main_fib population in C57BL/6 scRNA-seq dataset. (E, F) The main_fib cells labeled by stage and chamber.



Supplementary Fig. S5.

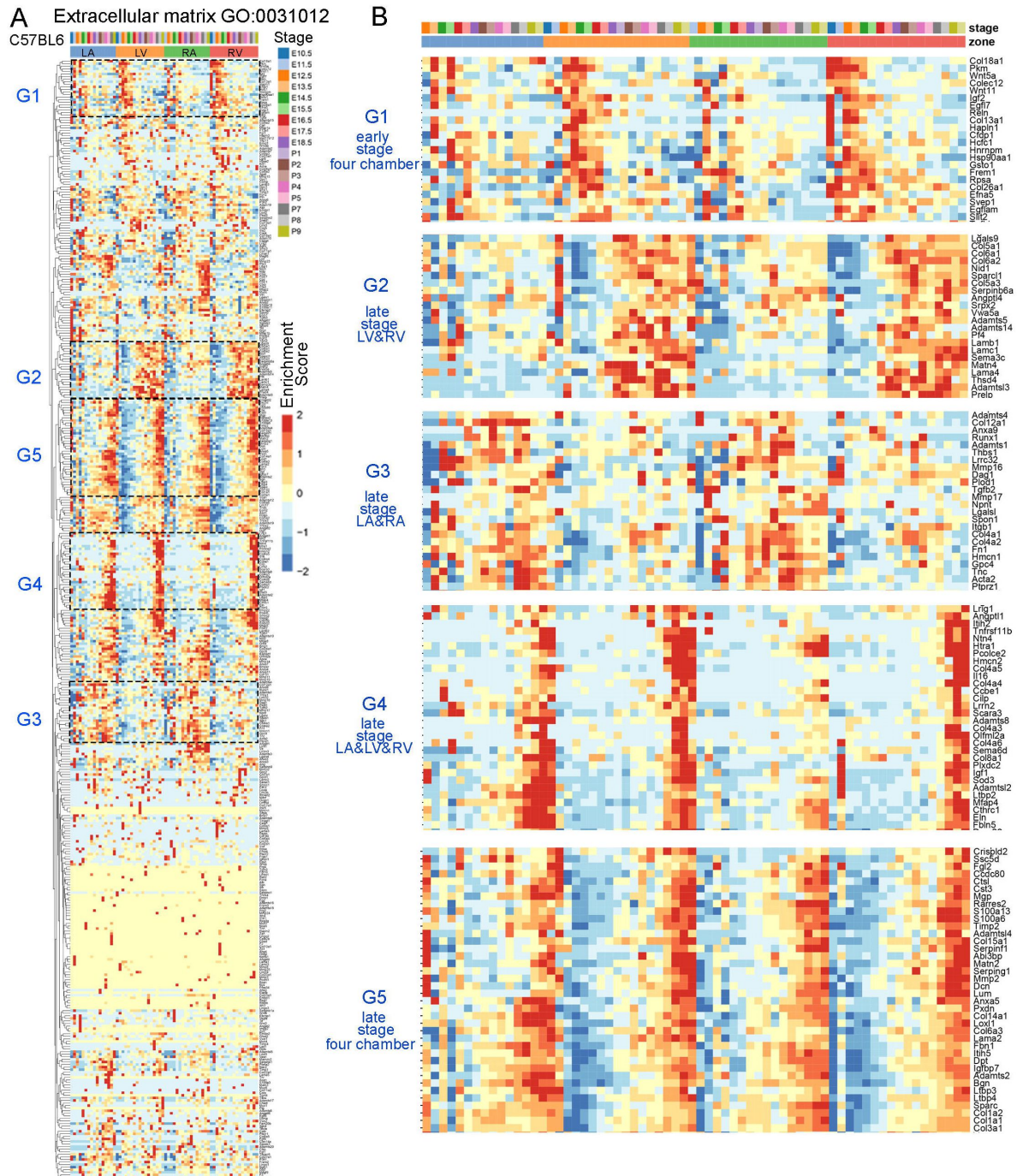
The expression pattern of extracellular matrix genes in the main population of cardiac fibroblasts.

(A) Clustering analysis of the ECM gene expression enrichments. (B) The group of genes (G1) that were highly expressed at early staged fibroblasts in all four zones. (C) The group of genes (G2) that were highly expressed in LV and RV at late embryonic and neonatal stages. (D) The genes (G3) that were preferentially expressed in LA and RA. (E) The genes (G4) that were highly expressed in LA, LV, and RV at neonatal stage.



Supplementary Fig. S6

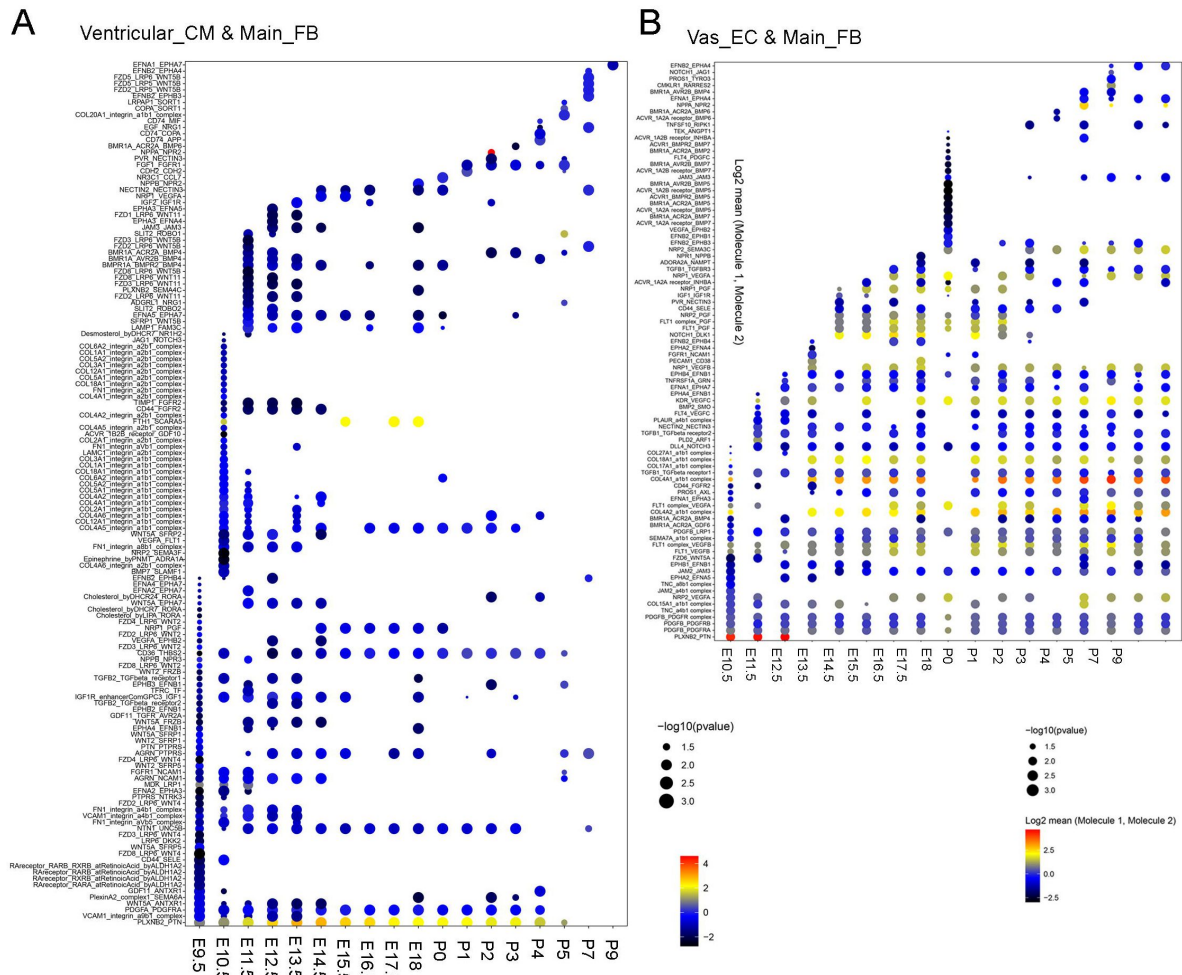
(A, B) The expression pattern of a group of genes (G5) that displayed expression in all four chambers of FBs (CD1 mice) at late embryonic and neonatal stages.



Supplementary Fig. S7

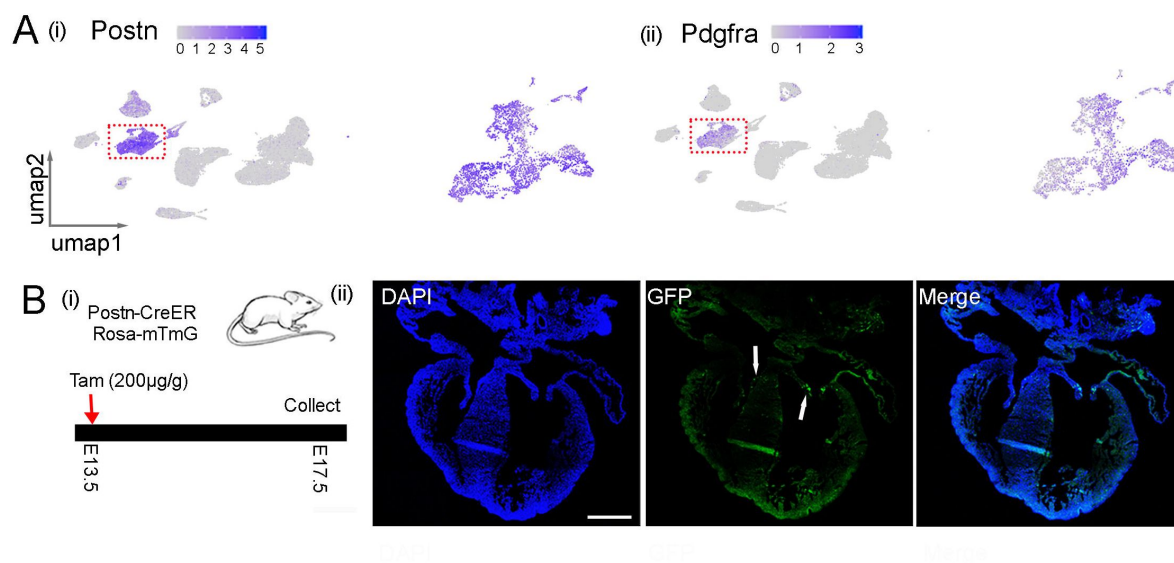
The expression pattern analysis of extracellular matrix genes in C57BL/6 FBs.

(A) Unsupervised clustering analysis of extracellular matrix genes expression in C57BL/6 FBs. (B) The groups of genes that display stage or zone-specific expression pattern.



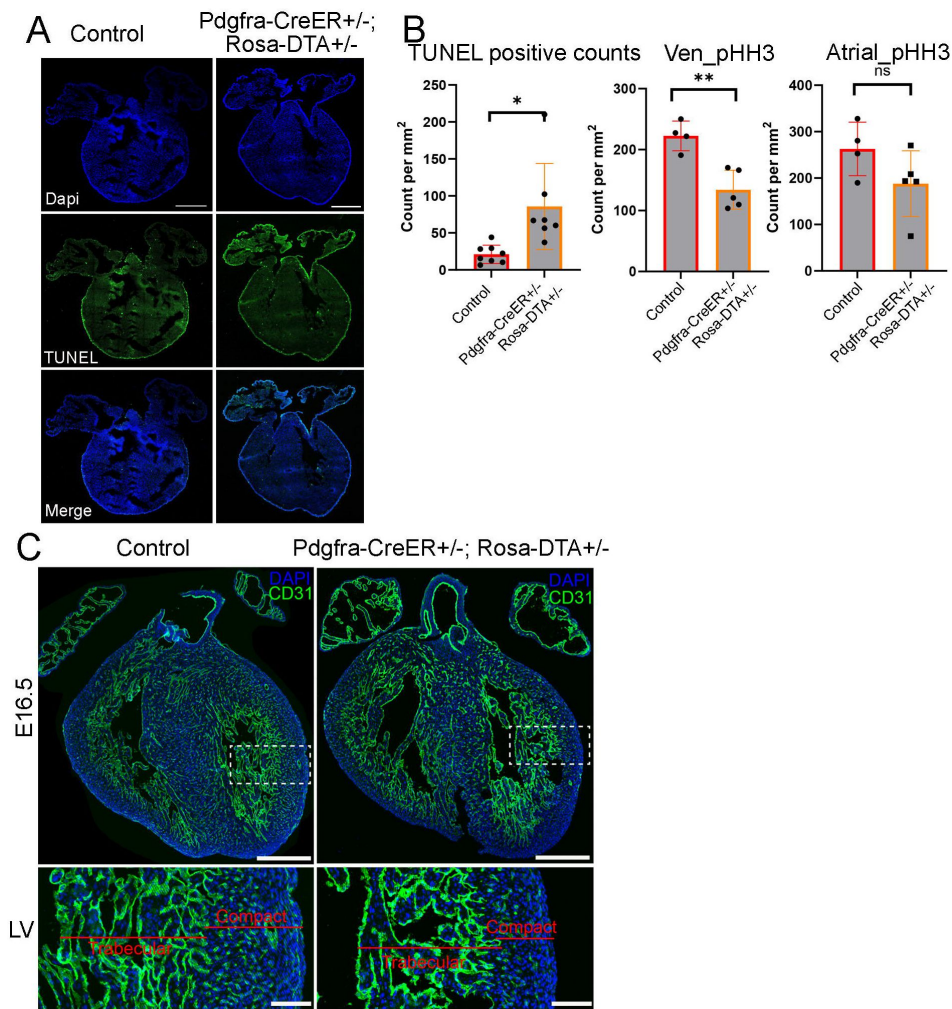
Supplementary Fig. S8

(A, B) The ligand-receptor interactions between FBs and ventricular CM or Vas_EC at each stage.



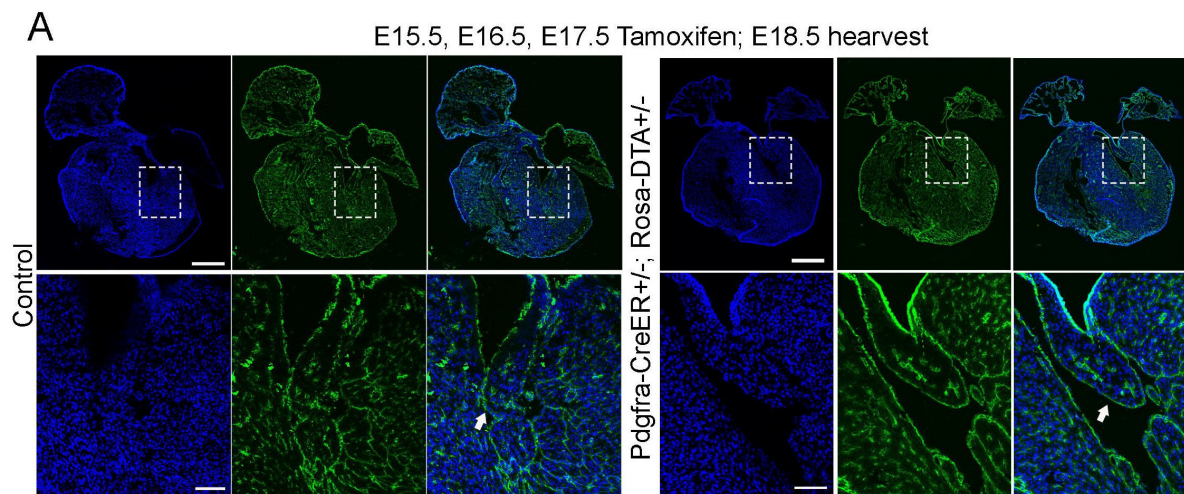
Supplementary Fig. S9

(Ai, ii) UMAP plots of *Postn* and *Pdgfra* expression in cardiac cells and cardiac fibroblasts. (B) (i) The workflow to lineage trace the *Postn*-CreER;mTmG labeled cells. (ii) A few cells in valves and septum were labeled in *Postn*-CreER; Rosa26-mTmG mouse hearts. Scale bar=500µm.



Supplementary Fig. S10

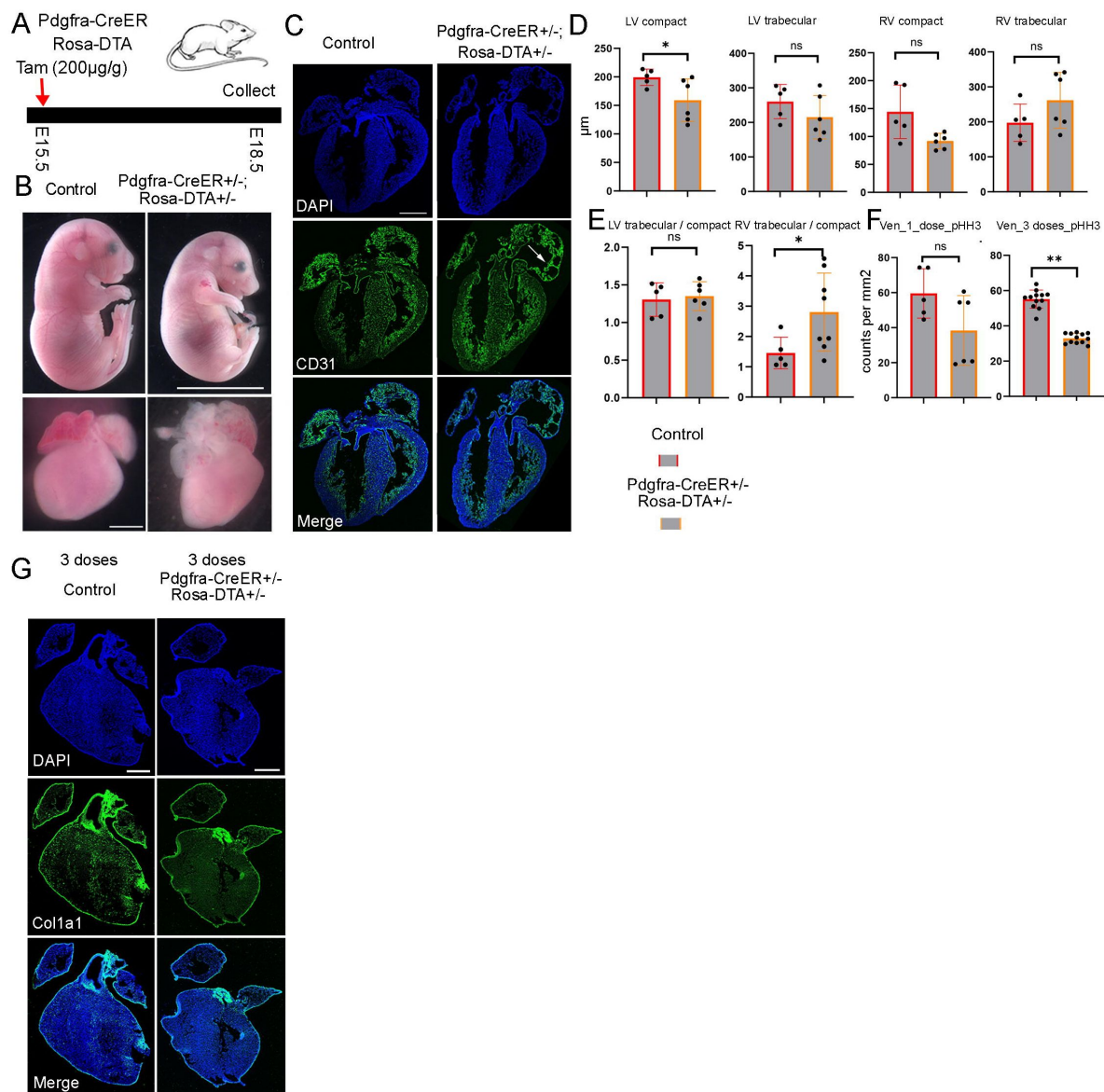
(A) Representative images of TUNEL staining in control and Pdgfra-CreER; Rosa-DTA hearts. (B) Quantification of TUNEL (7 sections from 2 ablated hearts and 8 sections from 2 control hearts) and pHH3 signals (5 sections from 2 ablated hearts and 4 sections from 2 control hearts). (C) Representative images illustrating the quantification of compact and trabecular myocardium thickness in control and ablated hearts at E16.5. Scale bar=500µm and 100µm in the whole heart images and the images with enlarged areas, respectively.



Supplementary Fig. S11

Comparative analysis of valve development in control and fibroblast-ablated hearts at E18.5.

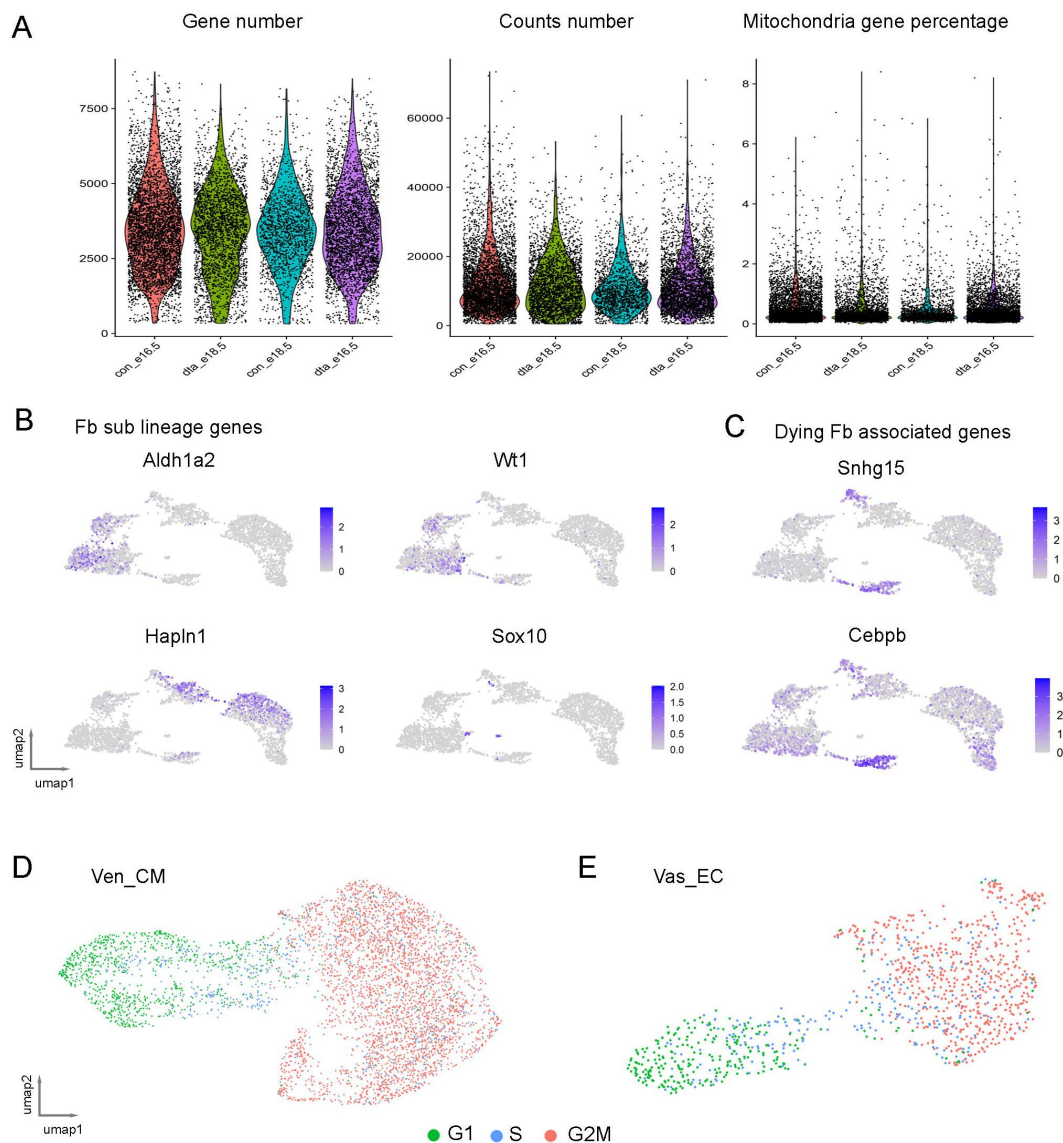
(A) Scale bar=500μm and 100μm in the whole heart images and the images with enlarged areas, respectively.



Supplementary Fig. S12

Quantification of the defects in control and ablated hearts with one dose of tamoxifen treatment at E15.5.

(A) Diagram of the experimental procedure with one dose of tamoxifen being administered at E15.5. (B) No obvious size differences were observed in the ablated embryos and hearts compared to controls. (C) CD31 staining analysis of control and ablated hearts. The littermate control mice were Pdgfra-CreER+/- (D). Quantification of the compact and trabecular myocardium thickness in LV and RV (6 sections from 2 ablated hearts and 5 sections from 2 control hearts). (E) Quantification of the ratio of trabecular to compact myocardium in LV and RV. (F) Quantification of pHH3 positive cells in control and ablated ventricles (6 sections from 2 ablated hearts and 5 sections from 2 control hearts for one dose of tamoxifen treated group, while 12 sections from 4 ablated hearts and 12 sections from 4 control hearts for three doses of tamoxifen treated group). (G) RNA staining analysis of *Col1a1* in control and ablated hearts with three doses of tamoxifen treatments. The *Col1a1* signal was dramatically reduced in the ablated hearts. * represents $p < 0.05$; ** represents $p < 0.01$. Scale bar=500µm.



Supplementary Fig. S13

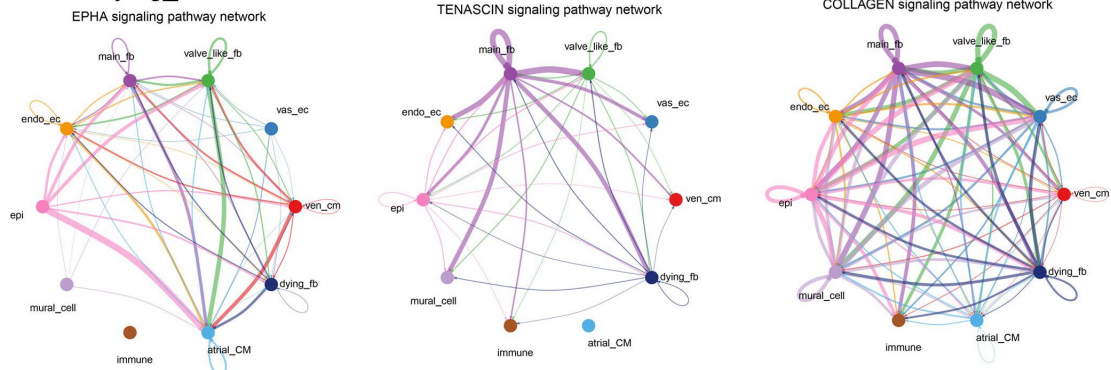
(A) QC plots of the filtered single cells including the total number of genes, total number of mRNA, and percentage of mitochondria genes. (B) The expression pattern of representative genes to identify fibroblast subpopulations. (C) The genes highly and specifically expressing in the dying_fb cells. (D, E) UMAP plots of Ven_CM and Vas_EC labeled by cell cycle phases.



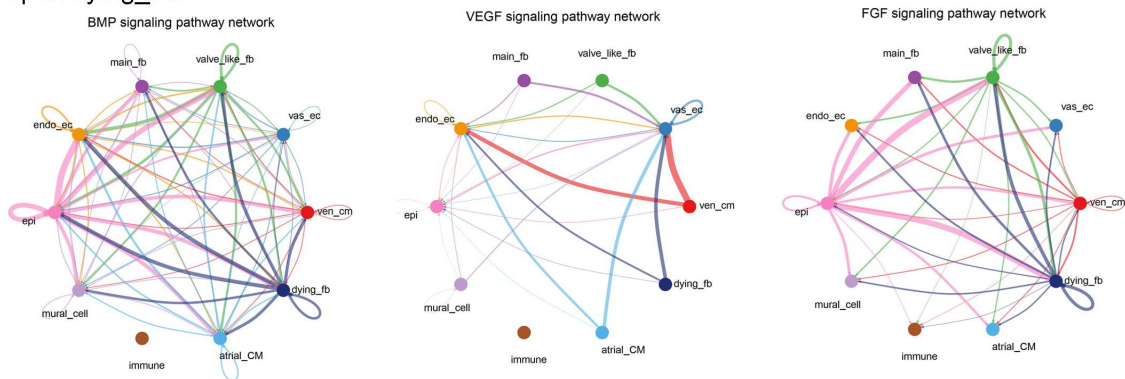
Supplementary Fig. S14

The predicted ligand-receptor interaction potentials between main_fb and Ven_CM or Vas_EC.

A Down in dying_Fb

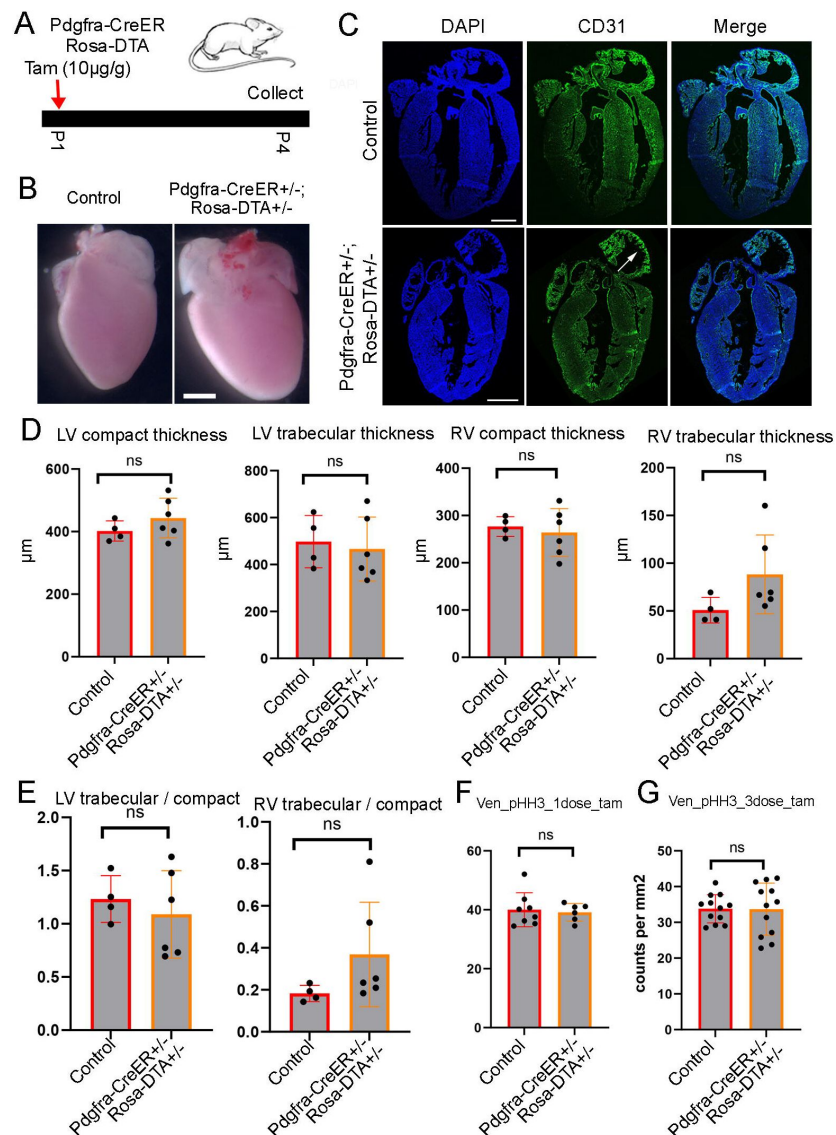


B Up in dying_Fb



Supplementary Fig. S15

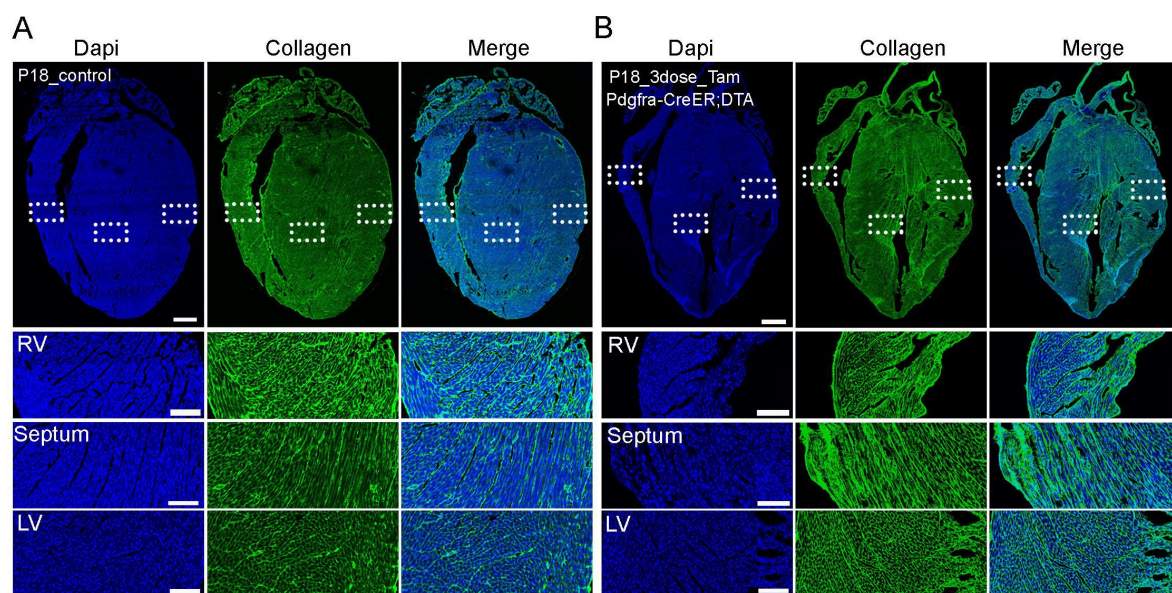
The signaling pathways that were found to reduce or increase their strengths in the dying_Fb compared to the main_fb and valve_like_fb.



Supplementary Fig. S16

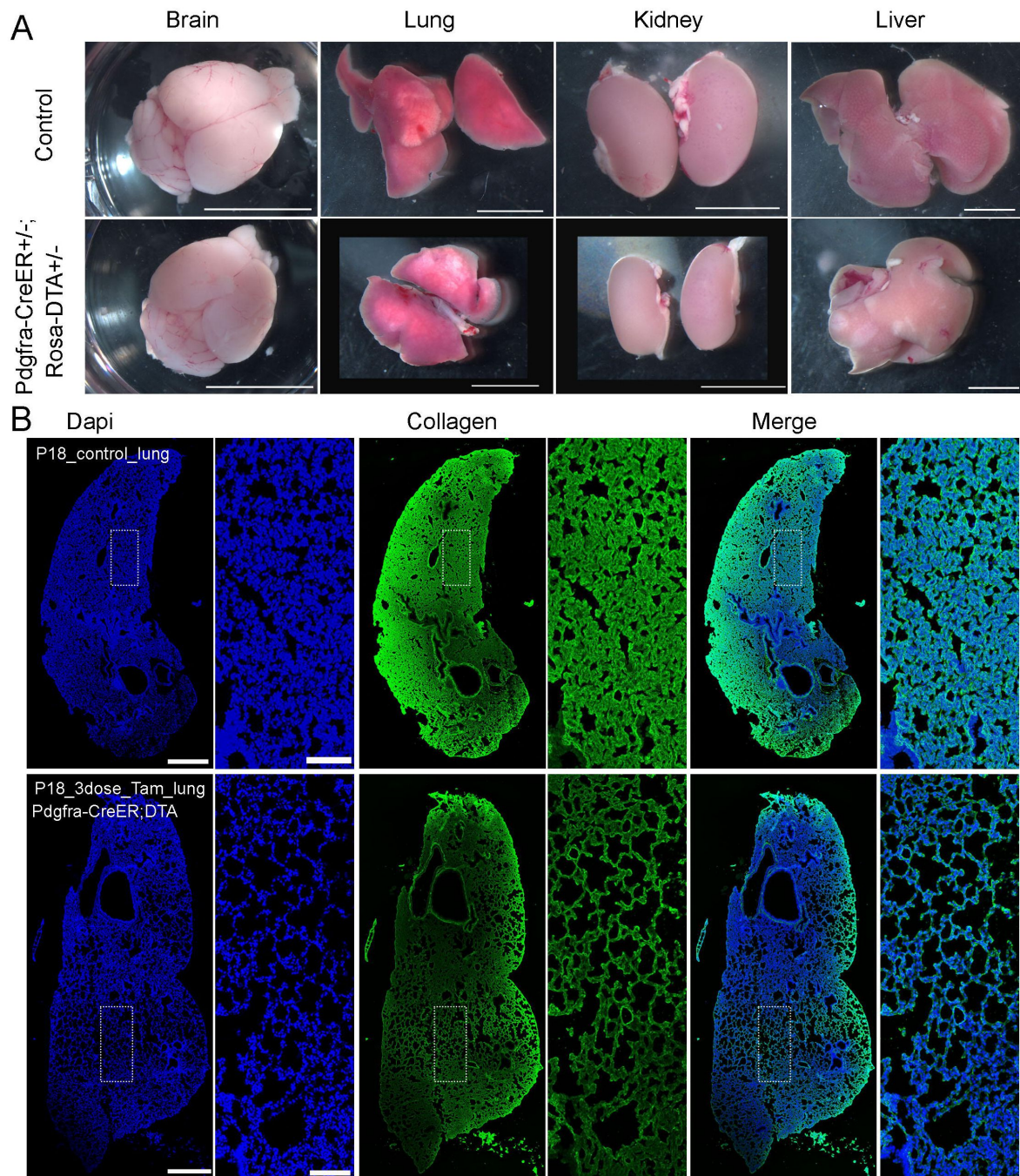
Quantification of the defects in control and ablated hearts at neonatal stage with one dose of tamoxifen treatment.

(A) Diagram of the experimental procedure with tamoxifen treatment at P1. (B) No obvious size differences were observed in the ablated hearts compared to control hearts. (C) Representative images of control and ablated hearts stained with CD31. The defects in the right atrium from ablated hearts were pointed out by an arrow. The littermate control mice include both double-negative and Pdgfra-CreER^{+/+} mice. Scale bar = 500µm. (D) Quantification of the compact and trabecular myocardium thickness in LV and RV (6 sections from 3 ablated hearts and 4 sections from 2 control hearts). (E) Quantification of the ratio of trabecular to compact myocardium in LV and RV after one dose of tamoxifen treatment (6 sections from 3 ablated hearts and 4 sections from 2 control hearts). (F) Quantification of pH3 positive cells in ventricular in control and ablated hearts after one dose of tamoxifen treatment (6 sections from 3 ablated hearts and 8 sections from 4 control hearts). (G) Quantification of pH3 positive cells in ventricular in control and ablated hearts after three doses of tamoxifen treatments (12 sections from 4 ablated hearts and 12 sections from 4 control hearts). * represents $p < 0.05$; ** represents $p < 0.01$.



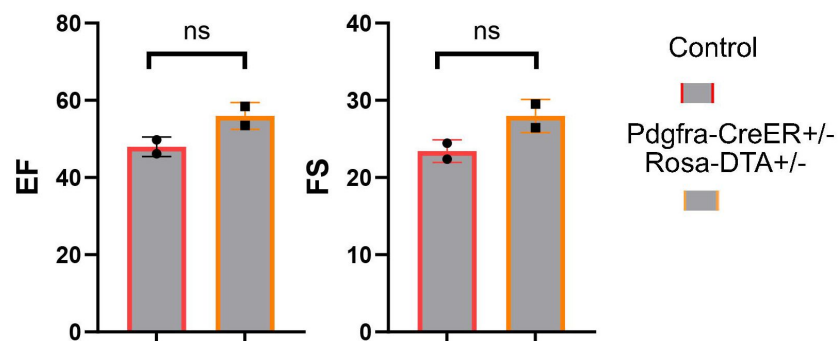
Supplementary Fig. S17

(A, B) Collagen accumulation in control and ablated hearts at P18. Scale bar=500µm and 150µm in the whole heart sections and enlarged sections, respectively.



Supplementary Fig. S18

(A) Representative images of brain, lung, kidney, and liver from control and Pdgfra-CreER; Rosa-DTA mice at P18 with tamoxifen treatment at P3 to P5. Scale bar = 5mm. (B) Collagen staining of lung from control and ablated mice at P18. The littermate control mice were Pdgfra-CreER^{+/-}. Scale bar=500μm and 150μm in the whole heart sections and enlarged sections, respectively.



Supplementary Fig. S19

Echocardiographic analysis of heart function in anesthetized control and ablated mice at P18.

The littermate control mice were Pdgra-CreER+/-.

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We'd like to thank all the members in the Li laboratory for their insightful discussions of this work. We are thankful for Dr. Wei Feng for his help on the MULTI-seq experiment. This research was supported in part by the University of Pittsburgh Center for Research Computing, RRID:SCR_022735, through the resources provided. Specifically, this work used the HTC cluster, which is supported by NIH award number S10OD028483.

Additional information

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Author Contributions

Y.D., Y.H., G.L. designed the experiments; Y.D. analyzed the ablated hearts with different assays; J.X. bred the mice, treated them with tamoxifen, and harvested the hearts from embryonic and neonatal mice. Y.H. performed the RNAScope and collagen staining experiments, Y.H. and G.L. performed the PLISH experiments; Y.D. performed the scRNA-seq experiment. H.T. and G.L. analyzed the scRNA-seq data; M.Z. and J.X. has performed the Echo scanning and data analysis. Y.D., Y.H., H.T., and G.L. prepared the manuscript; All the authors edited the manuscript.

Additional files

Supplemental Table 1. *Primer sequences for PLISH probes* [↗](#)

Supplemental Table 2. *The interaction strength values among the different cell types in Figure 2A.* [↗](#)

Supplemental Table 3. *The enrichment score of extracellular matrix genes in CD1 main population fibroblasts.* [↗](#)

Supplemental Table 4. *The enrichment score of extracellular matrix genes in C57BL/6 main population fibroblasts.* [↗](#)

Supplemental Table 5. *The list of ligand-receptor interactions between the main population of fibroblast and ventricular cardiomyocytes in CD1 dataset.* [↗](#)

Supplemental Table 6. *The list of ligand-receptor interactions between the main population of fibroblast and Vas_EC in CD1 dataset.* [↗](#)

Supplemental Table 7. *The list of genes that differentially expressed in control and ablated Ven_CM at e16.5 stage.* [↗](#)

Supplemental Table 8. *The list of genes that differentially expressed in control and ablated Ven_CM at e18.5 stage.* [↗](#)

Supplemental Table 9. *The list of genes that differentially expressed in control and ablated Vas_EC.* [↗](#)

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Reviewer #1 (Public review):

Summary:

The study by Deng et al reports single cell expression analysis of developing mouse hearts and examines the requirements for cardiac fibroblasts in heart maturation. The work includes extensive gene expression profiling and bioinformatic analysis. The prenatal fibroblast ablation studies show new information on the requirement of these cells on heart maturation before birth.

The strengths of the manuscript are the new single cell datasets and comprehensive approach to ablating cardiac fibroblasts in pre and postnatal development in mice. Extensive data are presented on mouse embryo fibroblast diversity and morphology in response to fibroblast ablation. Histological data support localization of major cardiac cell types and effects of fibroblast ablation on cardiac gene expression at different times of development.

A weakness of the study is that the major conclusions regarding collagen signaling and heart maturation are based on gene expression patterns and are not functionally validated.

<https://doi.org/10.7554/eLife.102305.2.sa3>

Reviewer #2 (Public review):

This study aims to elucidate the role of fibroblasts in regulating myocardium and vascular development through signaling to cardiomyocytes and endothelial cells. This focus is significant, given that fibroblasts, cardiomyocytes, and vascular endothelial cells are the three primary cell types in the heart. The authors employed a Pdgfra-CreER-controlled diphtheria toxin A (DTA) system to ablate fibroblasts at various embryonic and postnatal stages, characterizing the resulting cardiac defects, particularly in myocardium and vasculature development. Single-cell RNA sequencing (scRNA-seq) analysis of the ablated hearts identified collagen as a crucial signaling molecule from fibroblasts that influences the development of cardiomyocytes and vascular endothelial cells.

This is an interesting manuscript; however, there are several major issues, including an over-reliance on the scRNA-seq data, which shows inconsistencies between replicates.

Some of the major issues are described below.

(1) The CD31 immunostaining data (Figure 3B-G) indicate a reduction in endothelial cell numbers following fibroblast deletion using PdgfraCreER^{+/+}; RosaDTA^{+/+} mice. However, the scRNA-seq data show no percentage change in the endothelial cell population (Figure 4D). Furthermore, while the percentage of Vas_EC_s decreased in ablated samples at E16.5, the results at E18.5 were inconsistent, showing an increase in one replicate and a decrease in another, raising concerns about the reliability of the RNA-seq findings.

(2) Similarly, while the percentage of Ven_CM_s increased at E18.5, it exhibited differing trends at E16.5 (Fig. 4E), further highlighting the inconsistency of the scRNA-seq analysis with the other data.

(3) Furthermore, the authors noted that the ablated samples had slightly higher percentages of cardiomyocytes in the G1 phase compared to controls (Fig. 4H, S11D), which aligns with the enrichment of pathways related to heart development, sarcomere organization, heart tube

morphogenesis, and cell proliferation. However, it is unclear how this correlates with heart development, given that the hearts of ablated mice are significantly smaller than those of controls (Figure 3E). Additionally, the heart sections from ablated samples used for CD31/DAPI staining in Figure 3F appear much larger than those of the controls, raising further inconsistencies in the manuscript.

(4) The manuscript relies heavily on the scRNA-seq dataset, which shows inconsistencies between the two replicates. Furthermore, the morphological and histological analyses do not align with the scRNA-seq findings.

(5) There is a lack of mechanistic insight into how collagen, as a key signaling molecule from fibroblasts, affects the development of cardiomyocytes and vascular endothelial cells.

(6) In Figure 1B, Col1a1 expression is observed in the epicardial cells (Figure 1A, E11.5), but this is not represented in the accompanying cartoon.

(7) Do the *Pdgfra*^{CreER}^{+/-}; *Rosa*^{DTA}^{+/-} mice survive after birth when induced at E15.5, and do they exhibit any cardiac defects?

<https://doi.org/10.7554/eLife.102305.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors investigated fibroblasts' communication with key cell types in developing and neonatal hearts, with focus on critical roles of fibroblast-cardiomyocyte and fibroblast-endothelial cells network in cardiac morphogenesis. They tried to map the spatial distribution of these cell types and reported the major pathways and signaling molecules driving the communication. They also used Cre-DTA system to ablate *Pdgfra* labeled cells and observed myocardial and endothelial cell defects at development. They screened the pathways and genes using sequencing data of ablated heart. Lastly they reported a compensatory collagen expression in long term ablated neonate heart. Overall, this study provides us with important insight on fibroblasts' roles in cardiac development and will be a powerful resource for collagens and ECM focused research.

Strengths:

The authors utilized good analyzing tools to investigate on multiple database of single cell sequencing and Multi-seq. They identified significant pathways, cellular and molecular interactions of fibroblasts. Additionally, they compared some of their analytic findings with human database, and identified several groups of ECM genes with varying roles in mice.

Weaknesses:

This study is majorly based on sequencing data analysis. At the bench, they used very strident technique to study fibroblast functions by ablating one of the major cell population of heart. Also, experimental validation of their analyzed downstream pathways will be required eventually.

<https://doi.org/10.7554/eLife.102305.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The study by Deng et al reports single-cell expression analysis of developing mouse hearts and examines the requirements for cardiac fibroblasts in heart maturation. Much of this work is overlapping with previous studies, but the single-cell gene expression data may be useful to investigators in the field. The significance and scope of new findings are limited and major conclusions are largely based on correlative data.

Strengths:

The strengths of the manuscript are the new single-cell datasets and comprehensive approach to ablating cardiac fibroblasts in pre and postnatal development in mice.

Weaknesses:

There are several major weaknesses in the analysis and interpretation of the results.

(1) The major conclusions regarding collagen signaling and heart maturation are based on gene expression patterns and are not functionally validated. The potential downstream signaling pathways were not examined and known structural contributions of fibrillar collagen to heart maturation are not discussed.

We thank the reviewer for the comment. In this study, we mainly focused on the functional analysis of fibroblasts in heart development at embryonic and neonatal stages by using cell ablation system and single cell mRNA sequencing analysis. The further functional analysis of collagen pathway is interesting but out of the scope of this study. We will continue this line of research and share the results in the future. Moreover, through the analysis of single cell mRNA-sequencing data, we have predicted the downstream genes that are regulated by the collagen pathway in Fig 5C. We have also added sentences to highlight the structural role of collagen in affecting the related heart developmental processes.

(2) The heterogeneity of fibroblast populations and contributions to multiple structures in the developing heart are not well-considered in the analysis. The developmental targeting of fibroblasts will likely affect multiple structures in the embryonic heart and other organs. Lethality is described in some of these studies, but additional analysis is needed to determine the effects on heart morphogenesis or other organs beyond the focus on cardiomyocyte maturation being reported. In particular, the endocardial cushions and developing valves are likely to be affected in the prenatal ablations, but these structures are not included in the analyses.

We thank the reviewer for the comment. We have included a new figure presenting the fibroblast heterogeneity in developing hearts (Fig S3). We have also compared the valve structural differences at E18.5 (Fig S11).

(3) ECM complexity and extensive previous work on specific ECM proteins in heart development and maturation are not incorporated into the current study. Different types of collagen (basement membrane Col4, filamentous Col6, and fibrillar Col1) are known to be expressed in fibroblast populations in the developing heart and have been studied extensively. Much also has been reported for other ECM components mentioned in the current work.

We thank the reviewer for the comment. We agree that the ECM is complex, and the functions of many of its components have been previously reported, as mentioned in the introduction.

In this study, our focus is to analyze the spatial and temporal expression patterns of various ECM genes in fibroblasts throughout developmental progression (Fig. S5–7). To further acknowledge previous work, we have added additional sentences and cited relevant literature on the role of collagen genes in developing hearts (page 4).

Reviewer #2 (Public review):

This study aims to elucidate the role of fibroblasts in regulating myocardium and vascular development through signaling to cardiomyocytes and endothelial cells. This focus is significant, given that fibroblasts, cardiomyocytes, and vascular endothelial cells are the three primary cell types in the heart. The authors employed a Pdgfra-CreER-controlled diphtheria toxin A (DTA) system to ablate fibroblasts at various embryonic and postnatal stages, characterizing the resulting cardiac defects, particularly in myocardium and vasculature development. scRNA-seq analysis of the ablated hearts identified collagen as a crucial signaling molecule from fibroblasts that influences the development of cardiomyocytes and vascular endothelial cells. This is an interesting manuscript; however, there are several major issues, including an over-reliance on the scRNA-seq data, which shows inconsistencies between replicates. Some of the major issues are described below.

The comments are the same as the comments for “Recommendations for the authors”. Please see the responses below.

Reviewer #3 (Public review):

The authors investigated fibroblasts' communication with key cell types in developing and neonatal hearts, with a focus on the critical roles of fibroblast-cardiomyocyte and fibroblast-endothelial cell networks in cardiac morphogenesis. They tried to map the spatial distribution of these cell types and reported the major pathways and signaling molecules driving the communication. They also used Cre-DTA system to ablate Pdgfra labeled cells and observed myocardial and endothelial cell defects at development. They screened the pathways and genes using sequencing data of ablated hearts. Lastly, they reported compensatory collagen expression in long-term ablated neonate hearts. Overall, this study provides us with important insight into fibroblasts' roles in cardiac development and will be a powerful resource for collagens and ECM-focused research.

Strengths:

The authors utilized good analyzing tools to investigate multiple databases of single-cell sequencing and Multiseq. They identified significant pathways and cellular and molecular interactions of fibroblasts. Additionally, they compared some of their analytic findings with a human database, and identified several groups of ECM genes with varying roles in mice.

Weaknesses:

This study is majorly based on sequencing data analysis. At the bench, they used a very strident technique to study fibroblast functions by ablating one of the major cell populations of the heart. Considering the importance of the fibroblast population, intriguing in vivo findings were expected. Also, they analyzed the downstream genes in ablated hearts, but did not execute any experimental validation for any of the targets.

Recommendations for the authors:

Reviewing Editor Comments:

All three reviewers found the large amount of scRNA-Seq data compelling and valuable, and they noted that the study's conclusions based on the scRNA Seq and fibroblast ablating align closely with previously published studies. Therefore, a more thorough discussion and integration of the current findings with prior studies are recommended. Each reviewer provided specific feedback to improve the manuscript, correct errors, and strengthen the overall presentation, and please edit the manuscript accordingly. Additionally, further validation of the scRNA-Seq data through more data analysis, reference comparisons, or additional experiments is encouraged.

Reviewer #1 (Recommendations for the authors):

(1) The heterogeneity of fibroblasts and ECM components in the developing heart needs to be considered in the analysis and description of results. There are extensive reports in both of these areas that would inform the gene expression and ablation studies being reported.

We thank the reviewer for the comment. We have added a supplemental figure (Fig. S3) analyzing the heterogeneity of fibroblasts during development and described the results on page 3 and 4. Through the analysis of single-cell mRNA sequencing data, we identified four distinct populations of fibroblasts and further performed RNA scope to examine their spatial locations. Additionally, we agree with the reviewer that there are many types of ECM components, which we have addressed in the introduction (page 2). Furthermore, we have conducted a detailed analysis of the spatial and temporal expression patterns of ECM genes throughout developmental progression (Figs. S5–7).

(2) One of the novel aspects of the work is the prenatal ablation of cardiac fibroblasts. Embryonic lethality was observed in some cases, but the specific cardiac structural anomalies or potential vascular effects were not described. The contributing role of cardiac fibroblasts to valvuloseptal development, which was likely affected in these studies, was not described.

We thank the reviewer for the comment. Since the heart sections were not initially prepared to compare valve differences between control and ablation conditions, most sections do not include valve structures. However, in the small subset of sections that do contain valves, we have compared valve structures in control and ablated hearts at E18.5 following three doses of tamoxifen treatment from E15.5 to E17.5. In mutants, the valves appear shorter compared to controls. Specifically, we observed that in control hearts, the mitral valve was already connected to the papillary muscle, whereas in ablated hearts, the valve leaflet at similar position was not. We have included these images as a new supplemental figure (Fig. S11). Regarding vascular defects, we have described them in Fig. 3C and 3F.

(3) The major conclusions regarding collagen signaling and heart development are based on correlations in gene expression and are not validated by functional data. What are the downstream signaling pathways affected and are they affected during development or with ablation? The main conclusions of the study do not take into account well-known structural functions of collagen in the developing heart.

We thank the reviewer for the comment. Through regulatory prediction analysis, we identified the collagen ligands Col1a1, Col5a1, and Col4a1 from the collagen family (Fig. 5C), which regulate multiple genes in cardiomyocytes, including Masp1. Masp1 is a member of the lectin complement pathway and potentially regulates cardiomyocyte migration during development. These collagen ligands also regulate multiple mitochondria-related genes, such

as Etfa, Ndubf10, Ndubs6, and Slc25a4, which are potentially important for cardiomyocyte development and maturation. Moreover, we agree with the reviewer that collagen is an important structural ECM protein, and its deletion or reduction could cause heart developmental defects due to its structural role. We have added a discussion on this possibility (page 8).

(4) The postnatal ablation studies are very similar to studies with the same mouse lines reported by Kurabara et al 2022 in JMCC (PMID 35569524) which came to similar conclusions and was not cited in the current work.

We thank the reviewer for the comment and apologize for overlooking this study. We have now included the citation on page 8.

(5) The discussion of a regenerative response with DTA ablation of fibroblasts is confusing. Proliferation was examined in cardiomyocytes which lose their regenerative capacity after birth in mice. However, cardiac fibroblasts can proliferate in response to injury throughout life which is not really a regenerative process.

We appreciate the reviewer's comment. To avoid confusion, we have replaced the term "regeneration" with "response to cell loss" and "compensation."

(6) Some of the descriptions of single-cell expression data are overstated (Page 7). Regulatory interactions, signaling pathway activation, or function cannot be determined from gene expression data alone.

We thank the reviewer for the comment. We agree that these conclusions rely on results from multiple assays. We have weakened the description of the analysis by emphasizing that the findings are predictive results from scRNA-seq analysis.

(7) In the last paragraph of the discussion "data not shown" should be shown or this information should be deleted. As written, the discussion does not present a clear description of what major new findings are being reported or why they are significant. The new insights into heart development are not specified.

We thank the reviewer for the comment. We have added the data as a supplemental figure (Fig. S19). Since this paragraph is part of the discussion, we believe the results are not conclusive at this stage and require further research to explore the potential protective role of fibroblast ablation in neonatal hearts.

Minor comments.

(1) Figure legends are missing information needed to understand what is being shown. For example, in Figure 2, collagen is visualized using CHP staining.

Thanks. We have gone through all figure legends to ensure that all necessary information has been provided.

(2) The hearts in Figure S15 are upside down.

Thanks. We have updated the figure.

(3) In Figure S16A, "brian" should be "brain".

Thanks. We have updated it.

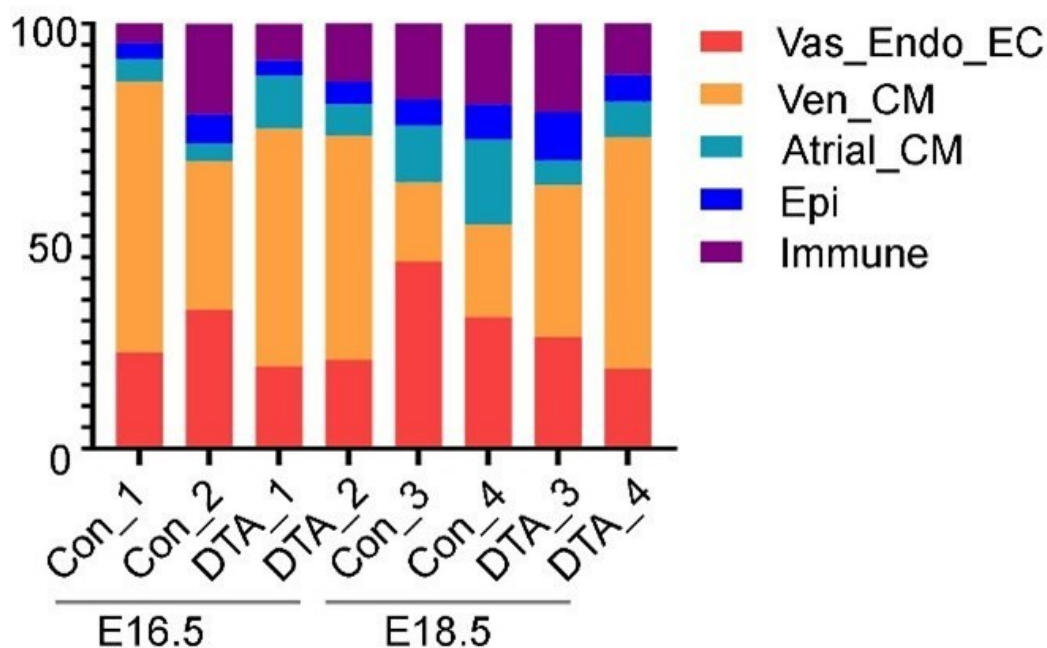
Reviewer #2 (Recommendations for the authors):

This is an interesting manuscript; however, there are several major issues, including an overreliance on the scRNA-seq data, which shows inconsistencies between replicates. Some of the major issues are described below.

*(1) The CD31 immunostaining data (Figures 3B-G) indicate a reduction in endothelial cell numbers following fibroblast deletion using *PdgfraCreER*^{+/-}; *RosaDTA*^{+/-} mice. However, the scRNA-seq data show no percentage change in the endothelial cell population (Figure 4D). Furthermore, while the percentage of Vas_EC decreased in ablated samples at E16.5, the results at E18.5 were inconsistent, showing an increase in one replicate and a decrease in another, raising concerns about the reliability of the RNA-seq findings.*

We thank the reviewer for the comment. We believe that measuring cell proportions in scRNA-seq results is sensitive and relies on a high number of total and target cells, similar to other cell counting assays such as FACS. As the reviewer pointed out, the proportions of Vas_EC in E18.5 replicates are inconsistent. Specifically, Col_4 at E18.5 showed a relatively low proportion of Vas_EC. Upon examining the cell numbers in each sample, we found that Col_4 had the lowest number of recovered cells, with approximately 760 in total, whereas the other samples had more than 920 cells each. Additionally, since immunofluorescence staining for CD31 marks both Vas_EC and Endo_EC, we combined these two cell types to increase the number of targeted cells. This analysis consistently showed that the ablated samples had lower proportions. However, given that the quantifications have also produced inconsistent results for other cell types, such as Ven_CM, as mentioned in the reviewer's next question, we have decided to delete this plot to avoid confusion.

Author response image 1.



(2) Similarly, while the percentage of Ven_CMs increased at E18.5, it exhibited differing trends at E16.5 (Figure 4E), further highlighting the inconsistency of the scRNA-seq analysis with the other data.

We thank the reviewer for the comment. Please see the response above.

(3) Furthermore, the authors noted that the ablated samples had slightly higher percentages of cardiomyocytes in the G1 phase compared to controls (Figures 4H, S11D), which aligns with the enrichment of pathways related to heart development, sarcomere organization, heart tube morphogenesis, and cell proliferation. However, it is unclear how this correlates with heart development, given that the hearts of ablated mice are significantly smaller than those of controls (Figure 3E). Additionally, the heart sections from ablated samples used for CD31/DAPI staining in Figure 3F appear much larger than those of the controls, raising further inconsistencies in the manuscript.

We thank the reviewer for the comment. We observed changes in G1-phase cardiomyocytes at both E16.5 and E18.5, with pathway enrichment primarily identified in E16.5 cardiomyocytes. At E16.5, the ablated hearts exhibited myocardial defects, including an increased trabecular-to-compact myocardium ratio and reduced vascular density. By E18.5, the ablated embryos had smaller hearts with reduced vascular density, although the trabecular-to-compact myocardium ratio showed no obvious changes. Regarding the larger section size in the ablated hearts compared to the control hearts, there are two reasons contributing to this discrepancy. First, the control and ablated heart sections have different scale bars. The ablated hearts were enlarged compared to control section. Secondly, the heart sections vary in size depending on their position. Sections taken from the middle of the heart are larger than those from the edges. In our initial comparison, we used an edge-positioned section from the control hearts and a middle-positioned section from the ablated hearts. To avoid confusion, we have now updated the control section to match the position of the ablated embryos more closely and used the same size of scale bars in the two images (Fig 3F).

(4) The manuscript relies heavily on the scRNA-seq dataset, which shows inconsistencies between the two replicates. Furthermore, the morphological and histological analyses do not align with the scRNA-seq findings.

We respectfully disagree with this comment from the reviewer. As shown in Figure 4B, the scRNAseq data from the two replicates are highly consistent. For inconsistencies in cell proportions and tissue section sizes, please refer to our responses above.

(5) There is a lack of mechanistic insight into how collagen, as a key signaling molecule from fibroblasts, affects the development of cardiomyocytes and vascular endothelial cells.

We thank the reviewer for the comment. In this study, we primarily focused on analyzing fibroblast function in heart development using cell ablation and single-cell mRNA sequencing. While further mechanistic analysis of the collagen pathway is intriguing, it falls outside the scope of this study. Additionally, our scRNAseq analysis identified multiple collagen ligands derived from fibroblasts that may regulate gene expression in Ven_CM and influence their development, as shown in Figure 5C. Although validating these predictions would be valuable, it is beyond the scope of this study. We will continue this line of research and share our findings in the future.

(6) In Figure 1B, Col1a1 expression is observed in the epicardial cells (Figure 1A, E11.5), but this is not represented in the accompanying cartoon.

We thank the reviewer for the comment. As stated in the main text (page 3), based on scRNA-seq and IF staining results, we observed that Col1a1 is also expressed in epicardial cells. In

the cartoon, we depicted the pattern of fibroblasts rather than Col1a1-positive cells, which is why we did not include epicardial cells.

(7) *What is the genotype of the control animals used in the study?*

We thank the reviewer for the comment. We have added the genotype information for the control embryos in the legends of the relevant figures.

(8) *Do the *Pdgfra*CreER^{+/-}; *Rosa*DTA^{+/-} mice survive after birth when induced at E15.5, and do they exhibit any cardiac defects?*

We thank the reviewer for the comment. This is an interesting question; however, we did not perform the experiment because administering tamoxifen to pregnant mice from E15.5 to E18.5 causes delivery complications, as reported in the literature (PMID: 23139287). Unfortunately, this prevents us from exploring this question further.

Reviewer #3 (Recommendations for the authors):

Overall, this is a comprehensive study substantiated by the evidence the authors provided in their findings. However, I have a few concerns to be addressed.

*(1) The claim by the authors that "at E17.5 and P3, each FB was in contact with approximately one Vas_EC and four CMs at both stages" is not fully convincing. RNA scope images for *Actn2* are not clear enough to lead the quantification (RNA scope images for *Cdh5* look better). I suggest performing imaging at higher magnification and the Z stack technique to provide a better understanding of their localization. Also, no changes in FBs adjacent cell numbers (CM&EC) with ages (P3) compared to E17.5? Any thoughts on the explanation?*

We thank the reviewer for the comment. We imaged the staining results using a confocal microscope at 20X resolution. We also considered imaging them at 40X; however, due to the large areas that need to be imaged in these sections, it was challenging to do so. Additionally, we identified each CM based on *Actn2* and DAPI staining information and are confident in the accuracy of our quantification results. Moreover, since each FB interacts with multiple CMs and Vas_EC in 3D projections, but our calculations are based only on 2D imaging sections, there may be discrepancies compared to a true 3D environment. We have added a sentence to address this limitation (page 9). Regarding the similar number of interactions observed at E17.5 and P3, we think there are two possibilities. First, the three cell types may proliferate in a synchronized manner, maintaining a consistent number of interactions. Second, these cell types may exhibit minimal proliferation during late embryonic and early neonatal stages. Instead, heart growth primarily occurs through CM hypertrophy, which does not significantly alter the number of interactions.

(2) *Fix the Capitalized font of RNA markers in Figure S2.*

Thanks. We have updated them.

(3) I appreciate the visualization of ligand-receptor interactions in collagen network comparison between FB to CM and FB to EC, and predictive analysis on the FB ligands that regulate differentially expressed genes in ablated heart CM and ECs.

We appreciate the reviewer for the comment.

*(4) The authors depleted *Pdgfra*-Cre cells at E10.5, and reported 100% DTA⁺ lethality after 3 days. Induction at E13.5 to ablate *Pdgfra*-Cre cells resulted in survival at least up*

to E16.5 age. What could be the possible reasons authors think that lead to embryo lethality when induced at E10.5? Did the authors analyze the expression of Pdgfra at E10.5 to E13.5 using Pdgfra antibody or Pdgfra-Cre labeling, or using the ScRNA seq data?

We thank the reviewer for the comment. The expression pattern of Pdgfra at E10.5 has been previously reported (PMID: 18297729) and shown to be highly expressed in the atrioventricular region, consistent with the Col1a1 expression pattern we profiled in this study. Therefore, we believe the embryonic lethality observed in the ablated embryos at E10.5 was likely due to the disruption of the atrioventricular structure. However, since Pdgfra is also expressed in other tissues at this stage, we cannot rule out the possibility that the ablation of non-cardiac tissues also contributed to the lethality.

(5) In terms of the findings on the trabeculation and compaction defects, please provide the images of the ventricles with markers to indicate the compact and trabecular zones and their defects.

Thanks! We have included images that illustrate the quantification of compact and trabecular myocardium thickness in control and ablated hearts (FigS10C).

(6) Did the author check the expression of any other marker for the vascular system in addition to CD31 to see the effects of ablated FB on coronary vasculature development?

We thank the reviewer for the comment. We analyzed only Cd31 to assess the effects of fibroblast ablation on the overall endothelial cell population. We did not separately examine the subpopulations, but this would be an interesting direction for future studies.

(7) Can the authors interpret how findings from PHH3 proliferation explain thinner compact and thicker trabeculae in ablated hearts?

We thank the reviewer for the comment and apologize for the misinterpretation of the results. We observed that the ablated hearts have a thinner compact myocardium, while the thickness of the trabecular myocardium remains unchanged, leading to an increased trabecular-to-compact myocardium ratio (Fig 3D). We have corrected the description in the manuscript accordingly. Moreover, since the compact myocardium has a higher proliferation rate than the trabecular myocardium, a reduction in overall cell proliferation is expected to have a more pronounced impact on the compact myocardium. Inhibition of compact myocardium proliferation has been reported to lead thinner compact myocardium and non-compaction defects (PMID: 31342111).

(8) The authors did not execute experiments to find the downstream target that causes compaction defects and endothelial cell density defects upon ablation of FBs. Can you project from your sequencing analysis what could be the potential downstream if you could execute bench-side experiments on this?

We appreciate the reviewer for the comment. We believe that the regulatory predictive results in Figures 5C and D from the scRNA-seq data analysis have provided a set of downstream candidates for validation. We could select some of the ligands, such as the collagen ligands Col1a1, Col4a1, and Col5a1, to treat the ablated embryos *in vivo* to assess whether they could partially rescue the myocardium defects. Additionally, we could conduct *ex vivo* experiments by co-culturing CM and FB, comparing them with CM alone and CM treated with the identified ligands. This would allow us to evaluate CM proliferation and the expression of downstream genes identified in the prediction results. However, as the reviewer suggested, these experiments are planned for future studies.

(9) Please provide the echocardiographic M mode images with a comparable number of cardiac cycles in control and ablated (Fig. 6H). Also, the heart rate of the ablated heart is too low to compare other parameters with the control. If you could stabilize the heart rate at comparable values to control the heart, it is possible that EF and FS values will be largely changed.

We thank the reviewer for the comment. As the echocardiographic analysis was performed on conscious mice, the lower heart rates in the ablated mice are a phenotype associated with the ablation. Unfortunately, we are unable to adjust them to the same as the control mice.

(10) Can you provide a numerical dataset for any one of the cell chat figures? Like in figure 2A, supporting the claim "However, in terms of interaction strength, FB exhibited the highest values compared to those of other cell types (Fig. 2A)".

Yes, we have added a supplemental table (Table S2) containing the numerical interaction weights. As shown in the table, the interactions between FB and other cell types have the highest values.

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